

Approximating Competitive Equilibrium by Nash Welfare*

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Abstract

We explore the relationship between two popular concepts in the allocation of divisible items: competitive equilibrium (CE) and allocations that maximize Nash welfare, i.e., allocations where the weighted geometric mean of the utilities is maximal. When agents have homogeneous concave utility functions, these two concepts coincide: the classical Eisenberg–Gale convex program that maximizes Nash welfare over feasible allocations yields a competitive equilibrium. However, these two concepts diverge for non-homogeneous utilities. From a computational perspective, maximizing Nash welfare amounts to solving a convex program for any concave utility functions, whereas computing CE becomes PPAD-hard already for separable piecewise linear concave (SPLC) utilities.

We introduce the concept of *Gale-substitute* utility functions, which is an analogue of the weak gross substitutes (WGS) property for the so-called Gale demand system. For Gale-substitutes utilities, we show that any allocation maximizing Nash welfare provides an approximate-CE with surprisingly strong guarantees, where every agent gets at least half the *maximum* utility they can get at any CE, and is approximately envy-free. Gale-substitutes include utility functions where computing CE is PPAD hard, such as all separable concave utilities and the previously studied non-separable class of Leontief-free utilities. We introduce a broad new class of utility functions called *generalized network utilities* based on the generalized flow model. This class includes SPLC and Leontief-free utilities, and we show that all such utilities are Gale-substitutes.

Conversely, although some agents may get much higher utility at a Nash welfare maximizing allocation than at a CE, we show a price of anarchy type result: for general concave utilities, every CE achieves at least $(1/e)^{1/e} > 0.69$ fraction of the maximum Nash welfare, and this factor is tight.

1 Introduction

Competitive (market) equilibrium is a central solution concept in economics to analyze and predict the outcomes of market interactions. The Fisher model is a fundamental market model where a set of n agents $\mathcal{A}$ arrive at the market with a fixed supply of divisible goods $\mathcal{M}$. Each agent i has a fixed budget b_i and preferences over bundles of goods given by a utility function $u_i(\cdot)$. At competitive equilibrium (CE), prices $\mathbf{p} \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ of goods and fractional allocation $\mathbf{x} \in \mathbb{R}_{\geq 0}^{\mathcal{A} \times \mathcal{M}}$ of goods to agents are such that each agent i gets an optimal bundle of goods, i.e., x_i maximizes $u_i(x_i)$ subject to $\langle \mathbf{p}, x_i \rangle \leq b_i$, and the market clears, i.e., the demand for each good meets its supply.

Equilibrium allocations satisfy many desirable properties, including Pareto optimality and the fairness guarantee of (weighted) envy-freeness. These properties make this an attractive concept also in other settings where no direct trade and money are involved. For example, competitive equilibrium with equal incomes (CEEI) is a popular mechanism in fair division, where a fictitious market is created by giving each agent the same budget [51]. We note that CEEI is a special case of the Fisher model, where $b_i = 1$ for each agent i .

It is standard to assume that the utility functions $u_i(\cdot)$ are monotone non-decreasing, concave, and satisfy $u_i(\mathbf{0}) = 0$. Classical results in mathematical economics established the existence of an equilibrium in the Fisher model as well as more general exchange market models, by Arrow and Debreu [4] and by McKenzie [44]. These results are based on fixed point theorems and thus do not lead to efficient computational methods. Finding an equilibrium by efficient computation has been an important topic in economics, optimization, and computer science over the past decades.

Eisenberg and Gale [26] gave a convex programming formulation for CE with linear utility functions $u_i(x) = \sum_{j \in \mathcal{M}} v_{ij} x_j$. This program amounts to maximizing the weighted Nash welfare, i.e., the weighted geometric mean of

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agents' utilities over feasible allocations. That is, any allocation maximizing $\prod_i u_i(x_i)^{b_i}$ (equivalently, $\sum_i b_i \log u_i(x_i)$) over feasible allocations gives a CE, where the dual solution yields the corresponding prices $\mathbf{p}$. Eisenberg [25] showed that the same convex program yields competitive equilibria more generally for concave utility functions that are positively homogeneous of degree one, i.e., $u_i(\alpha x_i) = \alpha u_i(x_i)$ for any $\alpha > 0$. This includes classical examples of constant elasticity of substitution (CES), Cobb–Douglas, and Leontief utility functions.

The Nash welfare takes its name due to its connection with the bargaining problem introduced by Nash [46]. Nash studied general cooperative bargaining problems where the set of possible outcomes is a convex set $K \subseteq \mathbb{R}^n$ with a disagreement point $d \in K$ and showed that maximizing the geometric mean of the excess utilities above the disagreement utility amounts to the only possible bargaining outcome that satisfies certain natural requirements. Taking K as the set of feasible allocations and $d = \mathbf{0}$, the Nash bargaining solution is precisely the Nash welfare maximizing allocation in this setting [52].

Weak gross substitutes (WGS) utility functions are another broad class where a CE can be computed in polynomial time. Several fast algorithms have been designed for computing equilibria with WGS utilities [7, 16, 17, 30]. While the WGS class includes many homogeneous utilities, there are also non-homogeneous examples for which the Eisenberg–Gale solution may not correspond to a CE.

WGS and homogeneous utility functions cover most known cases where a CE can be computed in polynomial time. Already for the seemingly simple case of separable piecewise concave utilities, computing a CE is PPAD-hard [14]. In fact, it is PPAD-hard even for bounded linear utilities, when there is an upper bound on the utility from each good, i.e., $u_i(x) = \sum_{j \in \mathcal{M}} \min\{v_{ij}x_j, \ell_{ij}\}$ [8]. Example A.1 shows that the set of CE can be non-convex and disconnected already for bounded linear utilities.

1.1 Our contributions In this paper, we study the relationship between two fundamental concepts in the allocation of divisible goods: competitive equilibrium and maximum Nash welfare allocations. We exhibit a close relationship between these solution concepts well beyond the homogeneous case. First, we show a ‘*price of anarchy*’ type result [24, 42]. One can interpret Nash welfare as a global measure of social utility. In contrast, a CE can be seen as an outcome reached by autonomous agents trying to improve their utilities individually. It turns out that, in the general setting of concave utilities, any CE recovers at least $(1/e)^{1/e} > 0.69$ fraction of the maximum Nash welfare, and this factor is tight (see Example A.4).

THEOREM 1.1. *For Fisher markets under concave utility functions, the Nash welfare at any CE is at least $(1/e)^{1/e}$ times the maximum Nash welfare.*

We then study guarantees of the maximum Nash welfare solution in the context of CE. A major advantage of maximum Nash welfare is efficient computability: it can be arbitrarily well approximated for any concave utility functions. In contrast, computing CE becomes PPAD-hard already in very simple cases. In this context, our first simple result shows that the maximum Nash welfare yields an approximate CE for all concave utility functions. By an α -demand-approximate CE, we mean allocations and prices such that every agent receives at least $1/\alpha$ -fraction of the maximum possible utility (see Definition 2.6). As in the case of the Eisenberg–Gale program, one can interpret the optimal Lagrangian multipliers as prices. The result also implies approximate envy-freeness.

THEOREM 1.2. *For Fisher markets under concave utility functions, any Nash welfare maximizing allocation $\mathbf{y}$, together with the dual prices $\mathbf{q}$, form a 2-demand-approximate CE. This also implies that $\mathbf{y}$ is half envy-free when b_i 's are identical, i.e., $u_i(y_i) \geq \frac{1}{2}u_i(y_k)$, for all $i, k \in \mathcal{A}$.*

Theorem 1.2 already shows that maximum Nash welfare provides remarkable fairness and efficiency guarantees for all concave utility functions. However, there can be multiple equilibria, giving vastly different utilities to an agent. In Example A.5, we show that there could be a gap of $\Omega(n)$ in the utility of some agent between two different competitive equilibria. Hence, an agent could be much better off in some CE than in the maximum Nash welfare solution.

As the main contribution of the paper, we show that maximum Nash welfare provides surprisingly strong guarantees for a broad class of utility functions. For this, we explore the *Gale demand correspondence* [30, 47]. For an agent $i \in \mathcal{A}$ with budget b_i and utility function $u_i(\cdot)$ at prices $\mathbf{q}$, the Gale demand correspondence comprises of the allocations y_i that maximize $b_i \log u_i(y_i) - \langle \mathbf{q}, y_i \rangle$. It follows from Lagrangian duality that in a Nash welfare maximizing solution, every agent receives an allocation from their Gale demand correspondence.

The reason behind the equivalence of CE and maximum Nash welfare for homogeneous utilities is that, for this class, the standard demand correspondence and the Gale demand correspondence coincide. Nevertheless, these two demands can

behave very differently for non-homogeneous utility functions, e.g., the Gale demand can be non-monotonic in the prices, whereas the standard demand is always monotonic (see Example A.3).

We identify a critical property of the Gale demand correspondence that we call *Gale-substitutes property* (Definition 2.10). This is analogous to the weak gross substitutes (WGS) property of the standard demand correspondence. In particular, it asserts that if some prices decrease then the demand for goods with unchanged prices may only decrease.

Competitive equilibrium becomes more subtle when the utility functions are *satiabile*: there is an absolute maximum value of the utility that can be reached, regardless of the prices—bounded linear utilities mentioned above are such an example. To accommodate satiable utilities, we also introduce a stronger variant of the definition, called the Σ -*Gale-substitutes property*. We note that for *non-satiabile* utilities, the Gale-substitutes and Σ -Gale-substitutes properties coincide. Figure 1 illustrates the containment relationships among various utility classes discussed in this paper, with those in which the Fisher market equilibrium can be efficiently computed highlighted in grey.

Our main result shows that for this class of utility functions, the utility of any agent at the Nash welfare maximizing allocation is at least half the *maximum* utility they may get at *any* CE.

THEOREM 1.3. *Consider a Fisher market instance where all utility functions are Σ -Gale-substitutes. Then, for any competitive equilibrium $(\mathbf{x}, \mathbf{p})$ and any Nash welfare maximizing allocation $\mathbf{y}$, $u_i(y_i) \geq \frac{1}{2}u_i(x_i)$ holds for every $i \in \mathcal{A}$.*

As a counterpart to the ‘*price of anarchy*’ result in Theorem 1.1, one could call this a ‘*price of authority*’ result. The agents can either accept an allocation proposed by a central authority on the principle of maximizing Nash welfare or converge to a market solution (using some dynamics). By accepting the centrally proposed solution, each agent may be worse off by at most a factor 2 compared to the best possible equilibrium outcome. On the other hand, outcomes at different competitive equilibria can be very different, and the possible outcomes can be difficult to predict given the computational hardness of equilibria. Hence, for (Σ) -Gale-substitutes utility functions, the Nash welfare maximizing allocation provides a guaranteed good outcome to all agents simultaneously while also being efficiently computable. Thus, Theorem 1.3 establishes a strong relationship between CE and maximum Nash welfare beyond homogeneous functions and provides a natural alternative to the *equilibrium selection* problem [5] in this setting where multiple equilibria are possible.

We call a utility function $u(\cdot)$ *separable* if $u(x) = \sum_{j \in \mathcal{M}} v_j(x_j)$. This class gives an important example of Σ -Gale-substitutes. Note that such utilities, in general, are neither homogeneous nor WGS. As mentioned above, competitive equilibrium computation is already PPAD-hard for the separable piecewise linear concave (SPLC) case when each v_j is piecewise linear. This class has attracted significant attention on its own [2, 12, 13, 31, 53].

THEOREM 1.4. *Every separable utility function is Σ -Gale-substitutes.*

We consider two further examples of piecewise linear Gale-substitutes utilities:

- *Leontief-free (LF) utilities* [28, 32, 37]: Consider a set K of segments, and let $u(x) \stackrel{\text{def}}{=} \max_{\sum_{k \in K} u_k} u_k = \sum_{j \in \mathcal{M}} a_{jk} z_{jk}$, $u_k \leq \ell_k$; $\forall k \in K$; $\sum_{k \in K} z_{jk} \leq x_j \forall j \in \mathcal{M}$; $\mathbf{z} \geq 0$. That is, an LF function is given by a set of linear functions u_k ’s with a maximum utility limit ℓ_k on each $k \in K$. LF captures interesting non-separable PLC functions, and includes SPLC as a special case where each segment $k \in K$ has exactly one j with $a_{jk} > 0$.
- *Generalized network utilities*: We introduce a new and broad class of utility functions defined by generalized flow instances. Here, the goods correspond to source nodes in a flow network with supply x_j , $j \in \mathcal{M}$. Each arc has a capacity and a gain factor, and the utility $u(x)$ is the maximum flow value that can reach a designated sink node t from supplies x . The precise definitions are given in Section 2.1. LF can be formulated as a special case of a network with two layers, the middle layer between the sources and the sink corresponding to the segments K .

We show the following theorem in our full version:

THEOREM 1.5. *Generalized network utility functions are Gale-substitutes.*

Generalized flows is a classical network flow model that can model commodity transportations with losses, as well as conversions between different types of entities such as in financial networks; see Chapter 15 in [1] for a description and various applications.

Theorem 1.3 already shows that in Fisher markets with non-satiabile generalized network utilities, Nash welfare maximizing allocations give a good approximation to utilities at competitive equilibrium. However, satiable generalized

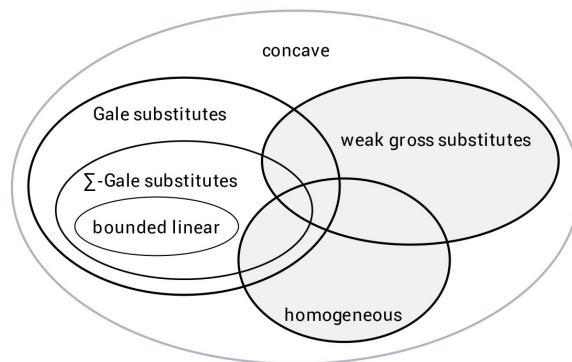


Figure 1: The graph illustrates the containment relationships among different utility classes, with the classes where the Fisher market equilibrium can be computed efficiently highlighted in grey.

network utilities may not satisfy the stronger Σ -Gale-substitutes property required in Theorem 1.3. Example A.10 shows that the theorem does not hold if only assuming the Gale substitutes property. For satiable utility functions, it is natural to look at *thrifty (strong) equilibria*, which are guaranteed to exist [33]. In such equilibria, each agent receives an optimal bundle at the cheapest possible price (see Definition 2.8). An appealing property of this concept is that thrifty equilibria are Pareto efficient, whereas this may not hold for (weak) competitive equilibrium. We also show that the statement of Theorem 1.3 remains true for generalized network utilities and for thrifty equilibria.

THEOREM 1.6. *Consider a Fisher market instance where all utility functions are generalized network utilities. Then, for any thrifty competitive equilibrium $(\mathbf{x}, \mathbf{p})$ and any Nash welfare maximizing allocation $\mathbf{y}$, $u_i(y_i) \geq \frac{1}{2}u_i(x_i)$ holds for every $i \in \mathcal{A}$.*

Finally, we note that the statement of Theorem 1.3 does not hold for arbitrary concave utilities. In Example A.8, we show that without the Gale-substitutes property, agents may get much higher utilities in CE than in the maximum Nash welfare allocations.

REMARK 1.1. *We note that no polynomial time approximation scheme (PTAS) can exist that, for some $\varepsilon > 0$, provides any agent at least $(1 - \varepsilon)$ times their maximum CE utility. This is due to the fact that, as shown in Example A.1, multiple thrifty equilibria may exist, each providing different utilities to the agents. The existence of an allocation that ε -approximates the best CE utility for each agent would, by compactness, imply the existence of an allocation where every agent achieves their best CE utility. However, this would contradict the fact that thrifty equilibria are Pareto efficient.*

1.2 Related work This paper strengthens the relationship between the two fundamental concepts of competitive equilibrium and maximum Nash welfare. Both are intensely studied concepts with a variety of applications, so we briefly mention further related results. As mentioned earlier, computing equilibria in Fisher markets is polynomial time for homogeneous and WGS utility functions, and beyond that, the problem is essentially PPAD-hard. For SPLC utilities, achieving a $\frac{1}{11}$ -approximate-CE is PPAD-hard [21]. We note that [14, 21, 53] use a different definition of approximate-CE, where every agent gets an optimal bundle, but the market clears approximately. Our approximate-CE definition also appears in several previous works, e.g., in [16, 17, 19].

Fisher markets are a special case of more general Arrow-Debreu (exchange) markets. Computing equilibria in exchange markets is polynomial time for WGS utility functions (see, e.g., [7, 17, 30]) and for CES utilities with parameter $-1 \leq \rho < 0$ [15], and beyond that the problem is essentially PPAD-hard [12], even for homogeneous functions, such as Leontief and CES with parameter $\rho < -1$ [13, 18].

The maximum Nash welfare is a classic objective for allocating goods to agents. It has received significant attention in the literature on social choice and fair division; see, e.g., [45]. It provides remarkable fairness guarantees also for indivisible goods: under additive valuations, maximum Nash welfare allocations attain the “unreasonable” fairness guarantee of envy-freeness up to one good (EF1) [10]. Furthermore, for more general subadditive valuations, the fairness guarantee of $\frac{1}{2}$ -approximate envy-freeness up to any good (EFX) can be achieved at a $\frac{2}{3}$ -approximate maximum Nash welfare allocation [27].

Another interesting line of research proposes Nash-bargaining-based solutions for allocation problems, offering efficient computation and other desirable structural properties; see, e.g., [52, 39, 34].

Overview The remainder of the paper is organized as follows. Section 2 defines all models and definitions. Section 3 provides a brief overview of the techniques used. Section 4 proves Theorem 1.1, showing the tight bound of $(1/e)^{1/e}$ on the efficiency of competitive equilibria. Section 5 presents several properties of the Gale demand correspondence and proves Theorem 1.2, showing that Nash welfare maximizing allocations yield 2-demand approximate-CE. Section 6 proves Theorem 1.3, showing the stronger guarantees of the maximum Nash welfare allocations for Gale-substitutes utility functions. Section 7 proves Theorem 1.4, showing that separable utilities are Σ -Gale substitutes. Several examples are deferred to the Appendix.

2 Preliminaries

Notation For a function $f : X \rightarrow Y$, let $\text{dom}(f) \subseteq X$ denote its domain. For $x \in \mathbb{R}^n$, let $\text{supp}(x) = \{i : x_i \neq 0\}$ denote its support. For $x, y \in \mathbb{R}^n$, we use $\langle x, y \rangle = x^\top y$ for the standard scalar product.

Subgradients and supergradients The functions we consider in this paper are not always differentiable. We will use sub- and supergradients, defined as follows.

DEFINITION 2.1. Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, a vector $\mathbf{g} \in \mathbb{R}^n$ is a supergradient of f at the point $\mathbf{x} \in \text{dom}(f)$ if for all $\mathbf{y} \in \text{dom}(f)$, $f(\mathbf{x}) + \langle \mathbf{g}, \mathbf{y} - \mathbf{x} \rangle \geq f(\mathbf{y})$. Similarly, $\mathbf{g} \in \mathbb{R}^n$ is a subgradient of f at $\mathbf{x} \in \text{dom}(f)$ if $f(\mathbf{x}) + \langle \mathbf{g}, \mathbf{y} - \mathbf{x} \rangle \leq f(\mathbf{y})$ for all $\mathbf{y} \in \text{dom}(f)$. We denote the set of all supergradients of f at $\mathbf{x}$ as $\partial^\wedge f(\mathbf{x})$, set of all subgradients as $\partial^\vee f(\mathbf{x})$.

Thus, $\mathbf{g}$ is a supergradient of f if and only if $-\mathbf{g}$ is a subgradient of $-f$. We recall that a function f is convex if and only if $\partial^\vee f(\mathbf{x}) \neq \emptyset$ for every $\mathbf{x} \in \text{dom}(f)$ and concave if and only if $\partial^\wedge f(\mathbf{x}) \neq \emptyset$ for every $\mathbf{x} \in \text{dom}(f)$. For a differentiable convex (concave) function, the unique subgradient (supergradient) at every point is the gradient $\nabla f(\mathbf{x})$.

2.1 Utility functions Given a set $\mathcal{M}$ of divisible goods, we will focus on utility functions with the following properties.

DEFINITION 2.2. A function $u(\cdot) : \mathbb{R}_{\geq 0}^{\mathcal{M}} \rightarrow \mathbb{R}$ is a utility function if $u(\mathbf{0}) = 0$, $u(\cdot)$ is concave, monotone non-decreasing, and there exists an $x \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ such that $u(x) > 0$.

We say that a utility function is *separable* if $u(x) = \sum_{j \in \mathcal{M}} v_j(x_j)$. We will also distinguish between satiable and non-satiable utilities.

DEFINITION 2.3. A utility function $u(\cdot)$ is non-satiable if and only if, for any $x \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$, one can find $x' \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$, such that $u(x') > u(x)$. Otherwise, we say the utility function is satiable.

Demand correspondence and Gale demand correspondence By a *price vector*, we mean a vector $\mathbf{p} \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$. Given prices $\mathbf{p}$ and a budget $b \in \mathbb{R}_{\geq 0}$, we define the *demand correspondence* as the set of maximum utility bundles achievable at $\mathbf{p}$ and b :

$$(2.1) \quad \mathcal{D}^u(\mathbf{p}, b) \stackrel{\text{def}}{=} \arg \max_{x \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} \{u(x) : \langle \mathbf{p}, x \rangle \leq b\}.$$

Note that if u is a non-satiable utility function, then $\langle \mathbf{p}, x \rangle = b$ must hold for every $x \in \mathcal{D}^u(\mathbf{p}, b)$.

$\mathcal{D}^u(\mathbf{p}, b)$ is always a non-empty compact set when $\mathbf{p} > 0$. However, in cases when some prices may be 0, the supremum of utility over the budget constraint might be ∞ , resulting in unbounded demand. This may happen even when the supremum value is finite, and causes technical difficulties. To circumvent these issues, we adopt the following assumption regarding utility functions $u(\cdot)$:

ASSUMPTION 1. For any $\mathbf{p} \geq \mathbf{0}$ and $b > 0$, if there is a sequence of allocations $\{x^{(k)}\}_{k=1}^\infty$ such that $\langle \mathbf{p}, x^{(k)} \rangle$ converges to b and $u(x^{(k)})$ converges to some finite value $M < \infty$, then there exists a finite allocation $x^* \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ such that $u(x^*) = M$ and $\langle \mathbf{p}, x^* \rangle \leq b$.

This assumption always holds if the set of possible allocations is restricted to a compact domain $X \subseteq \mathbb{R}_{\geq 0}^{\mathcal{M}}$; note [47] makes such a compactness assumption. For the unrestricted domain $\mathbb{R}_{\geq 0}^{\mathcal{M}}$, it can also be shown for arbitrary piecewise linear utility functions, in particular, for generalized network utilities defined below.

The *Gale demand correspondence* is defined as

$$(2.2) \quad \mathcal{GD}^u(\mathbf{p}, b) \stackrel{\text{def}}{=} \arg \max_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b \log u(y) - \langle \mathbf{p}, y \rangle.$$

We define the domain of $\mathcal{GD}^u$ with respect to prices as

$$(2.3) \quad \text{pdom}(u) \stackrel{\text{def}}{=} \left\{ \mathbf{p} \in \mathbb{R}_{\geq 0}^{\mathcal{M}} : \exists b > 0, \sup_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b \log u(y) - \langle \mathbf{p}, y \rangle < +\infty \right\}.$$

Under Assumption 1, $\mathbf{0} \in \text{pdom}(u)$ if and only if $u(\cdot)$ is satiable. Every strictly positive $\mathbf{p} > \mathbf{0}$ is in $\text{pdom}(u)$, and if u is satiable, then $\text{pdom}(u) = \mathbb{R}_{\geq 0}^{\mathcal{M}}$ (see Lemma 5.2). In Section 5, we show the following.

LEMMA 2.1. *Under Assumption 1, for any $\mathbf{p} \in \text{pdom}(u)$ and any $b > 0$, $\mathcal{GD}^u(\mathbf{p}, b) \neq \emptyset$.*

Gale demands may behave very differently from the standard demand correspondence. Example A.3 shows the counterintuitive property increasing the prices $\mathbf{p}$ may actually increase the Gale utility value $\max_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b \log u(y) - \langle \mathbf{p}, y \rangle$.

Recall the notions of bounded linear, separable piecewise linear (SPLC) and Leontief-free (LF) utilities from the Introduction. We now define generalized network utility functions.

Generalized network utilities First, we introduce the generalized flow model. A *generalized flow instance* is defined as follows. Let $G = (V, E)$ be a directed graph with a sink node $t \in V$, a supply set $S \subseteq V \setminus \{t\}$, a capacity vector $h \in \mathbb{R}_{\geq 0}^E$, and positive gain factors $\gamma \in \mathbb{R}_{> 0}^E$. We let $\partial^{\text{in}}(S)$ and $\partial^{\text{out}}(S)$ denote the set of arcs entering and leaving a subset $S \subseteq V$, respectively. For a node $v \in V$, we use $\partial^{\text{in}}(v) \stackrel{\text{def}}{=} \partial^{\text{in}}(\{v\})$ and $\partial^{\text{out}}(v) \stackrel{\text{def}}{=} \partial^{\text{out}}(\{v\})$. For a vector $f \in \mathbb{R}_{\geq 0}^E$, for every arc $e = (v, w)$, we interpret f_e as the flow entering e at node v . The flow gets multiplied by the gain factor γ_e while traversing the arc, with $\gamma_e f_e$ amount reaching node w . Thus, the *net flow* at node $v \in V$ is defined as

$$\text{net}_f(v) \stackrel{\text{def}}{=} \sum_{e \in \partial^{\text{in}}(v)} \gamma_e f_e - \sum_{e \in \partial^{\text{out}}(v)} f_e.$$

Given a *supply vector* $x \in \mathbb{R}_{\geq 0}^S$, we say that $f \in \mathbb{R}_{\geq 0}^E$ is a *feasible generalized flow with supply x* and capacity h if $0 \leq f \leq h$, and $\text{net}_f(v) \geq -x_v$ for all $v \in S$, and $\text{net}_f(v) \geq 0$ for all $v \in V \setminus (S \cup \{t\})$. The *maximum generalized flow value from supply x* is the maximum amount of $\text{net}_f(t)$ among all feasible flows with supply x . It corresponds to the optimum value of the following linear program:

$$\begin{aligned} & \max \text{net}_f(t) \\ & \text{net}_f(v) \geq -x_v \quad \forall v \in S \\ & \text{net}_f(v) \geq 0 \quad \forall v \in V \setminus (S \cup \{t\}) \\ & 0 \leq f \leq h. \end{aligned} \quad (\text{GenflowLP}(x))$$

DEFINITION 2.4. *The function $u : \mathbb{R}_{\geq 0}^{\mathcal{M}} \rightarrow \mathbb{R}_{\geq 0}$ is a generalized network utility function if $u(x)$ is the maximum generalized flow value from supply x in some generalized flow instance where the supply set is $S = \mathcal{M}$, the maximum flow amount is finite for any supply vector, $u(\mathbf{0}) = 0$, and there exists $x \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ with $u(x) > 0$.*

The following properties are immediate from the definition, and show that these are indeed utility functions in accordance with Definition 2.2.

PROPOSITION 2.1. *If $u : \mathbb{R}_{\geq 0}^{\mathcal{M}} \rightarrow \mathbb{R}_{\geq 0}$ is a generalized network utility function, then $u(x)$ is monotone non-decreasing and concave, and satisfies Assumption 1.*

SPLC and Leontief-free as special cases An SPLC function can be represented as a generalized network utility function by setting $V = \mathcal{M} \cup \{t\}$, and for each $j \in \mathcal{M}$, adding a capacitated arcs $e = (j, t)$ for each linear segment of v_j , where γ_e is the marginal utility on this segment and h_e is the length of the segment. Similarly, we can represent a Leontief-free utility as a generalized network utility function by setting $V = \mathcal{M} \cup K \cup \{t\}$; recall K is the set of segments. For each $j \in \mathcal{M}$ and $k \in K$, we add an arc $e = (j, k)$ with gain factor $\gamma_e = a_{jk}$ and capacity $h_e = \infty$. For each segment $k \in K$, we add an arc $e = (k, t)$ with $\gamma_e = 1$ and $h_e = \ell_k$.

Utilities from production Assume that agents are competing for access to m machines $\mathcal{M}$ that can produce ℓ different products. Machine j can produce a_{jk} units of product k per time unit. The revenue of product k is r_k . There are additionally limits t_{jk} on how much of each product can be produced on each machine by the agent, as well as a limit d_k on the maximum possible amount producible from k . Given x_j time on machine $j \in \mathcal{M}$, the utility of the agent is the maximum possible revenue subject to these constraints. This can be formulated using a two layer generalized flow network similarly to Leontief-free as above (albeit with additional capacity bounds t_{jk} .) The model allows for more complex production networks with possibly multiple layers corresponding to raw materials, intermediary and final products.

2.2 Competitive equilibrium In a *Fisher market instance*, given is a set of agents $\mathcal{A}$ and a set of divisible goods $\mathcal{M}$, with one unit available from each good. Agents have utility functions $u_i : \mathbb{R}_{\geq 0}^{\mathcal{M}} \rightarrow \mathbb{R}_{\geq 0}$ and budgets $b_i \in \mathbb{R}_{> 0}$, $i \in \mathcal{A}$; we denote the budget vector as $\mathbf{b} = (b_i)_{i \in \mathcal{A}}$. An allocation vector is $\mathbf{x} \in \mathbb{R}_{\geq 0}^{\mathcal{A} \times \mathcal{M}}$, where x_{ij} denotes the amount of good j allocated to agent i . A *price vector* is a vector $\mathbf{p} \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$. The demand correspondence of agent i will be denoted as $\mathcal{D}_i(\mathbf{p}, b_i)$.

DEFINITION 2.5. (COMPETITIVE EQUILIBRIUM) *Given a Fisher market instance as above, the allocations and prices $(\mathbf{x}, \mathbf{p})$ form a competitive (market) equilibrium if the following hold:*

- (i) *every agent gets an optimal utility at these prices: $x_i \in \mathcal{D}_i(\mathbf{p}, b_i)$ for every agent $i \in \mathcal{A}$*
- (ii) *no good is oversold: $\sum_i x_{ij} \leq 1$ for all $j \in \mathcal{M}$.*
- (iii) *every item with positive price is fully sold: $\sum_i x_{ij} = 1$ if $p_j > 0$.*

In a *demand-approximate competitive equilibrium*, agents may have suboptimal bundles:

DEFINITION 2.6. (APPROXIMATE COMPETITIVE EQUILIBRIUM) *Given a Fisher market instance and $\alpha \geq 1$, the allocations and prices $(\mathbf{x}, \mathbf{p})$ form a α -demand approximate competitive equilibrium if*

- (i) *$\langle \mathbf{p}, x_i \rangle \leq b_i$ for each $i \in \mathcal{A}$, and i gets at least α -fraction of the optimal utility at these prices: $u_i(x_i) \geq \frac{1}{\alpha} \max_{x_i \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} \{u_i(x_i) : \langle \mathbf{p}, x_i \rangle \leq b_i\}$.*
- (ii) *no good is oversold: $\sum_i x_{ij} \leq 1$ for all $j \in \mathcal{M}$.*
- (iii) *every item with positive price is fully sold: $\sum_i x_{ij} = 1$ if $p_j > 0$.*

It is a well-established fact that when all b_i 's are identical, the competitive equilibrium attains exact envy-freeness. Here, we demonstrate that our definition of approximate competitive equilibrium also results in approximate envy-freeness.

DEFINITION 2.7. *For a given allocation $\mathbf{x} \in \mathbb{R}_{\geq 0}^{\mathcal{A} \times \mathcal{M}}$, we say it is β -envy-free if, for any agents i and j , the condition $u_i(x_i) \geq \frac{1}{\beta} u_i(x_j)$ holds.*

LEMMA 2.2. *When b_i 's are identical, the allocation $\mathbf{x}$ of an α -approximate competitive equilibrium is also α -envy-free.*

Proof. This follows directly from $u_i(x_i) \geq \frac{1}{\alpha} \max_{x_i \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} \{u_i(x_i) : \langle \mathbf{p}, x_i \rangle \leq b_i\} \geq \frac{1}{\alpha} u_i(x_j)$. $\square$

Thrifty equilibrium In case there are satiable utility functions, $\langle \mathbf{p}, x_i \rangle < b_i$ may be possible in an (exact) competitive equilibrium. Hylland and Zeckhauser [40] showed that competitive equilibrium may not be Pareto efficient due to satiable utilities. Mas-Colell [43] strengthened the definition of a competitive equilibrium to require that each agent choose a least-cost optimal plan. This is also known as a *strong equilibrium*.

DEFINITION 2.8. (THRIFTY COMPETITIVE EQUILIBRIUM) *Given a Fisher market instance, the allocations and prices $(\mathbf{x}, \mathbf{p})$ form a thrifty competitive equilibrium if $(\mathbf{x}, \mathbf{p})$ is a competitive equilibrium, and moreover, for every $i \in \mathcal{A}$, the allocation x_i minimizes $\langle \mathbf{p}, x_i \rangle$ among all allocations $\mathcal{D}_i(\mathbf{p}, b_i)$.*

LEMMA 2.3. ([33, 38]) *Thrifty competitive equilibria always exist and are Pareto efficient.*

Note that in case all utilities are non-satiable, competitive equilibria and thrifty competitive equilibria coincide.

2.3 Nash welfare

DEFINITION 2.9. Given a Fisher market instance, the Nash welfare of an allocation $\mathbf{y} \in \mathbb{R}_{\geq 0}^{A \times \mathcal{M}}$ is the weighted geometric mean of their utilities, namely,

$$\left(\prod_{i \in \mathcal{A}} u_i(x_i)^{b_i} \right)^{\frac{1}{\sum_{i \in \mathcal{A}} b_i}}.$$

The allocation $\mathbf{y} \in \mathbb{R}_{\geq 0}^{A \times \mathcal{M}}$ maximizes the (weighted) Nash welfare, if its Nash welfare is maximal along all allocations with $\sum_{i \in \mathcal{A}} y_{ij} \leq 1$ for all $j \in \mathcal{M}$, $\mathbf{y} \geq \mathbf{0}$.

Note that the Nash welfare maximizing allocations are precisely the optimal solutions to Eisenberg–Gale convex program [26].

$$(EG) \quad \max \sum_{i \in \mathcal{A}} b_i \log u_i(y_i) \quad \text{s.t.} \quad \sum_{i \in \mathcal{A}} y_{ij} \leq 1, \quad \forall j \in \mathcal{M} \quad \text{and} \quad \mathbf{y} \geq \mathbf{0}.$$

The Lagrangian dual of (EG) can be written as

$$(2.4) \quad \varphi(\mathbf{q}) \stackrel{\text{def}}{=} \sum_{i \in \mathcal{A}} \max_{y_i \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} (b_i \log u_i(y_i) - \langle \mathbf{q}, y_i \rangle) + \sum_{j \in \mathcal{M}} q_j.$$

We define the domain $\text{dom}(\varphi) \stackrel{\text{def}}{=} \cap_{i \in \mathcal{A}} \text{pdom}(u_i)$ (see (2.3)). Thus, $\text{dom}(\varphi)$ is the set of nonnegative prices $\mathbf{q}$ such that $\varphi(\mathbf{q}) < \infty$. Strong duality holds and the dual optimum is taken according since Slater's condition holds. Consequently, the optimal Nash welfare value equals the dual optimum

$$(2.5) \quad \min_{\mathbf{q} \in \text{dom}(\varphi)} \varphi(\mathbf{q}).$$

This optimum is taken (see Lemma 6.2), and we refer to the optimal $\mathbf{q} \in \text{dom}(\varphi)$ as *Gale prices*. We note that these may not be unique. To justify this term, let us recall the definition of the Gale demand correspondence (2.2). The term corresponding to agent i in the definition of $\varphi(\mathbf{q})$ is the value of the assignments in the Gale demand correspondence of this agent at prices $\mathbf{q}$ and budget b_i . We use the notation $\mathcal{GD}_i(\mathbf{q}, b_i)$ for the Gale demand for the utility u_i , $i \in \mathcal{A}$.

2.4 Gale substitute utilities We now define *Gale substitute* and Σ -*Gale substitute* utilities, the crucial classes where we show that the Nash welfare maximizing allocation recovers at least half of the utility of every agent in any possible competitive equilibrium (see Theorem 1.3).

DEFINITION 2.10. The utility function $u(\cdot)$ is Gale-substitutes if it satisfies the following properties:

- (i) for any $b > 0$, any $\mathbf{q}, \mathbf{q}' \in \text{pdom}(u)$, and any $y \in \mathcal{GD}^u(\mathbf{q}, b)$, if $\mathbf{q} \geq \mathbf{q}'$, then there exists a $y' \in \mathcal{GD}^u(\mathbf{q}', b)$ such that $y'_j \leq y_j$ whenever $q'_j = q_j$;
- (ii) for any $b \geq b' > 0$ and $\mathbf{q} \in \text{pdom}(u)$, and for any $y \in \mathcal{GD}^u(\mathbf{q}, b)$, there exists a $y' \in \mathcal{GD}^u(\mathbf{q}, b')$ such that $y' \leq y$.

We say that $u(\cdot)$ is Σ -Gale-substitutes, if it is Gale-substitutes and the following additional property holds:

$$(\Sigma) \quad \text{if } \mathbf{0} \in \text{pdom}(u), \quad b > 0 \text{ and } \mathbf{q} \in \mathbb{R}_{\geq 0}^{\mathcal{M}}, \text{ then for any } y \in \mathcal{GD}(\mathbf{0}, b), \exists y' \in \mathcal{GD}^u(\mathbf{q}, b) : y' \leq y.$$

REMARK 2.1. As noted above, $\mathbf{0} \in \text{pdom}(u)$ if and only if u is satiable; thus, (Σ) is vacuously true for non-satiable utilities. In the satiable case, it asserts that if y is a bundle of maximum possible utility, then at any prices there is an optimal bundle below y . Observe that this is immediate for separable utilities. On the other hand, (Σ) fails for the simple Leontief-free utility $u(x) = \max\{x_1 + x_2, 1\}$: $y = (1, 0)$ is an optimal bundle, whereas for $b = 1$, $\mathbf{q} = (2, 1)$, the unique solution in $\mathcal{GD}^u(\mathbf{q}, b)$ is $y' = (0, 1)$.

Next, we demonstrate a utility function that exhibits Gale-substitutes but not substitutes properties.

EXAMPLE 2.1. Consider an agent with budget b and a utility function over two goods given by $u(x) = \min\{x_1, 0.4\} + x_2$. This utility is separable and, as such, Gale-substitutes according to Lemma 7.1 shown later. However, we show that this is not weak gross substitutes (WGS): there exist prices $\mathbf{p} \leq \mathbf{p}'$ and $x \in \mathcal{D}^u(\mathbf{p}, b)$ such that for some good j with $p_j = p'_j$, every $x' \in \mathcal{D}^u(\mathbf{p}, b)$ has $x'_j < x_j$. That is, let $\mathbf{p} = (0.8, 1)$ and $\mathbf{p}' = (0.9, 1)$, and $b = 1$. It is easy to see that $x = (0.4, 0.68)$ is the unique optimal bundle at prices $\mathbf{p}$, and $x' = (0.4, 0.64)$ is the unique optimal bundle at prices $\mathbf{p}'$ in contradiction to WGS property.

3 Overview of techniques

The ‘price of anarchy’ result Theorem 1.1 is proved using an elementary argument on concavity and the log-sum inequality. The proof is inspired by an argument in Theorem 1 in [9]. Example A.4 shows that the bound $(1/e)^{1/e}$ is tight.

To prove Theorem 1.2 showing that the Nash welfare maximizing allocation is 2-demand-approximate CE, the key ingredient is Lemma 5.1, showing that the solutions in the Gale demand system are within budget, that is, for $y \in \mathcal{GD}^u(\mathbf{q}, b)$ we have $\langle \mathbf{q}, y \rangle \leq b$. Since at the Nash welfare maximizing solution, every agent gets an allocation from the Gale demand system at the dual prices $\mathbf{q}$, this lemma guarantees that the allocation is feasible. The 2-demand-approximate property then follows by a supergradient argument.

The proof of Theorem 1.3 relies on two key lemmas derived using Lagrangian duality and supergradient arguments. The first is a simple observation, showing that the Gale demand solution for lower prices yields at least half the utility of the utility maximizing bundle at the current prices.

LEMMA 3.1. *Consider a utility function $u : \mathbb{R}_{\geq 0}^M \rightarrow \mathbb{R}_{\geq 0}$, prices $\mathbf{p}, \mathbf{q} \in \text{pdom}(u)$ such that $\mathbf{q} \leq \mathbf{p}$. Then, $u(y) \geq \frac{1}{2}u(x)$ for any $y \in \mathcal{GD}^u(\mathbf{q}, b)$ and $x \in \mathcal{D}^u(\mathbf{p}, b)$.*

The second lemma shows that for every allocation of goods in the demand correspondence, there is a smaller or equal allocation of goods in the Gale demand correspondence. Although the first lemma was true for any utilities, this one critically relies on assuming the Σ -Gale-substitutes property.

LEMMA 3.2. *Let $u(\cdot)$ be a Σ -Gale-substitutes utility and let $x \in \mathcal{D}^u(\mathbf{p}, b)$ for $\mathbf{p} \in \mathbb{R}_{\geq 0}^M$ and $b > 0$. Then, $\mathbf{p} \in \text{pdom}(u)$, and there exists $y \in \mathcal{GD}^u(\mathbf{p}, b)$ such that $y \leq x$.*

To derive Theorem 1.3, we start with a competitive equilibrium $(\mathbf{x}, \mathbf{p})$, and show that the dual of the Eisenberg–Gale program (2.5), $\varphi(\mathbf{q})$, has an optimal solution $\mathbf{q}^*$ such that $\mathbf{q}^* \leq \mathbf{p}$. The statement then follows from Lemma 3.1.

An intuitive argument would be the following: we start from $\mathbf{q} = \mathbf{p}$ where the total Gale demand is less than the supply (by Lemma 3.2), and use a tâtonnement-like procedure using the Gale-substitutes property. As long as $\mathbf{q}$ is not a minimizer of $\varphi(\cdot)$, there will be an underdemanded item j with respect to Gale demands. Reducing p_j for such a good j may only decrease the Gale demand for all agents on all other goods. This iterative process converges to an equilibrium point and, since the prices are only decreasing, we can conclude that the optimal limit price vector satisfies $\mathbf{q}^* \leq \mathbf{p}$.

Rather than using this continuous procedure, we establish $\mathbf{q}^* \leq \mathbf{p}$ directly using Lagrangian duality. For any price vector $\mathbf{q}$, we get a subgradient in $\partial^\vee \varphi(\mathbf{q})$ from the difference between the supply and the total Gale demand. Further, for $\mathbf{q} \leq \mathbf{p}$ we get that if $q_j = p_j$ then the j -th component of this subgradient at $\mathbf{q}$ is non-negative. This is shown in Lemma 6.1, by making use of an allocation $\mathbf{y}$ derived using Lemma 3.2 for the prices $\mathbf{q}$. Thus, in the region $\{\mathbf{q} \mid 0 \leq \mathbf{q} \leq \mathbf{p}\}$, the subgradients on the boundary all point outward. Consequently, if $\mathbf{q}^* \not\leq \mathbf{p}$, then we can take the projection $\mathbf{q}$ of $\mathbf{q}^*$ to the boundary of the region, where this subgradient shows that $\mathbf{q}$ is also optimal.

In Section 7, we prove Theorem 1.4, i.e., every separable utility function is Σ -Gale-substitutes. We show the theorem for general, not necessarily differentiable utility functions using subgradient arguments. We next outline the argument under the simplifying assumption that utilities are differentiable and strictly concave. Let $u(y) = \sum_{j \in \mathcal{M}} v_j(y_j)$, and let $\nabla_j u(y) = \partial v_j(y_j)/\partial y_j$ denote the j -th component of the gradient. Then, the KKT conditions yield $\nabla_j u(y) \leq q_j u(y)/b$, with equality whenever $y_j > 0$ (see Lemma 7.2). Let $y \in \mathcal{GD}^u(\mathbf{q}, b)$ and $y' \in \mathcal{GD}^u(\mathbf{q}', b)$ for $\mathbf{q}' \leq \mathbf{q}$. Using the optimality conditions, it is easy to derive that $u(y') \geq u(y)$. Consequently, for every good j with $q_j = q'_j$ and $y_j > 0$, $\nabla_j u(y') \geq \nabla_j u(y)$ follows. By strict concavity, we must have $y'_j \leq y_j$ as required. The proof without the differentiability and strict concavity assumptions poses additional difficulties: for example, we would not be able to conclude $y'_j \leq y_j$ at this point, and might need a specific choice of y'_j from $\mathcal{GD}^u(\mathbf{q}', b)$.

Theorem 1.5 shows that generalized network utilities are Gale substitutes. To verify the Gale substitutes property, we need to take two price vectors $\mathbf{q} \geq \mathbf{q}'$ and $y \in \mathcal{GD}^u(\mathbf{q}, b)$, and construct a $y' \in \mathcal{GD}^u(\mathbf{q}', b)$ such that $y'_j \leq y_j$ whenever $q'_j = q_j$; we have a second, analogous construction for budget changes. The proof is a finite algorithm that transforms the optimal generalized flow for demands y to y' such that we throughout maintain primal-dual slackness condition corresponding to the Gale demand correspondence. The algorithm builds on the literature on maximum generalized flow algorithms. In particular, we use ideas from the classical highest gain augmenting path algorithm by Onaga [49]. While this does not give a polynomial-time algorithm, it was the starting point of further such algorithms, including [36, 48, 54].

4 Efficiency of competitive equilibria

We now show the ‘price of anarchy’ result Theorem 1.1.

THEOREM 4.1. (RESTATEMENT OF THEOREM 1.1) *Given a Fisher market instance, consider any competitive equilibrium $(\mathbf{x}, \mathbf{p})$ and a Nash welfare maximizing allocation $\mathbf{y}$. Then,*

$$\left(\prod_{i \in \mathcal{A}} u_i(x_i)^{b_i} \right)^{\frac{1}{\sum_{i \in \mathcal{A}} b_i}} \geq \left(\frac{1}{e} \right)^{\frac{1}{e}} \left(\prod_{i \in \mathcal{A}} u_i(y_i)^{b_i} \right)^{\frac{1}{\sum_{i \in \mathcal{A}} b_i}}.$$

Proof. Without loss of generality, we can assume $\sum_{i \in \mathcal{A}} b_i = 1$. Note that since $(\mathbf{x}, \mathbf{p})$ is a competitive equilibrium, then $u_i(x_i) \geq \min \left\{ 1, \frac{b_i}{\langle \mathbf{p}, y_i \rangle} \right\} u_i(y_i)$, as agent i can use their budget, b_i , to buy at least $\frac{b_i}{\langle \mathbf{p}, y_i \rangle}$ fraction of y_i . The lower bound holds because if $b_i < \langle \mathbf{p}, y_i \rangle$, then $u_i \left(\frac{b_i}{\langle \mathbf{p}, y_i \rangle} y_i \right) \geq \frac{b_i}{\langle \mathbf{p}, y_i \rangle} u_i(y_i)$ using concavity and $u_i(\mathbf{0}) = 0$. Thus,

$$\sum_{i \in \mathcal{A}} b_i \log \frac{u_i(y_i)}{u_i(x_i)} \leq \sum_{i \in \mathcal{A}} b_i \log \max \left\{ 1, \frac{\langle \mathbf{p}, y_i \rangle}{b_i} \right\}.$$

Let $b'_i \stackrel{\text{def}}{=} \langle \mathbf{p}, y_i \rangle$, and let $\mathcal{A}'$ be the set of agents such that $b'_i \geq b_i$, $B \stackrel{\text{def}}{=} \sum_{i \in \mathcal{A}'} b_i$, and $B' \stackrel{\text{def}}{=} \sum_{i \in \mathcal{A}'} b'_i$. Therefore, using the log sum inequality,

$$\sum_{i \in \mathcal{A}} b_i \log \frac{u_i(y_i)}{u_i(x_i)} \leq \sum_{i \in \mathcal{A}'} b_i \log \frac{b'_i}{b_i} \leq B \log \frac{B'}{B}.$$

By definition, $B \leq \sum_{i \in \mathcal{A}} b_i = 1$. We also note that $B' \leq \sum_{i \in \mathcal{A}} b'_i = \sum_{i \in \mathcal{A}} \langle \mathbf{p}, y_i \rangle \leq \sum_{j \in \mathcal{M}} p_j \leq \sum_{i \in \mathcal{A}} b_i = 1$.

$$\sum_{i \in \mathcal{A}} b_i \log \frac{u_i(y_i)}{u_i(x_i)} \leq \max_{B \leq 1} B \log \frac{1}{B} \leq \frac{1}{e}.$$

The statement follows. $\square$

5 Properties of the Gale demand correspondence

In this section, we show some properties of the Gale demand correspondence, and also derive Theorem 1.2.

LEMMA 5.1. *Given any utility function $u : \mathbb{R}_{\geq 0}^{\mathcal{M}} \rightarrow \mathbb{R}_{\geq 0}$, $b > 0$ and $\mathbf{q} \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$, for any $y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ such that $\langle \mathbf{q}, y \rangle > b$, there exists $y' \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ such that $\langle \mathbf{q}, y' \rangle \leq b$ and $b \log u(y') - \langle \mathbf{q}, y' \rangle > b \log u(y) - \langle \mathbf{q}, y \rangle$. In particular, if $y \in \mathcal{GD}^u(\mathbf{q}, b)$, then, $\langle \mathbf{q}, y \rangle \leq b$.*

Proof. Let $y' \stackrel{\text{def}}{=} y \cdot \frac{b}{\langle \mathbf{q}, y \rangle}$. By concavity and $u(\mathbf{0}) = 0$, we get $u(y') \geq \frac{b}{\langle \mathbf{q}, y \rangle} u(y)$. Thus,

$$b \log u(y') - \langle \mathbf{q}, y' \rangle \geq b \log u(y) - b \log \frac{\langle \mathbf{q}, y \rangle}{b} - b > b \log u(y) - \langle \mathbf{q}, y \rangle,$$

where the second inequality used $\log(\alpha) + 1 < \alpha$ for $\alpha > 1$. $\square$

Next, we derive some technical properties of the price domain $\text{pdom}(u)$.

LEMMA 5.2. *For any utility function $u : \mathbb{R}_{\geq 0}^{\mathcal{M}} \rightarrow \mathbb{R}_{\geq 0}$ and $b > 0$, the following properties of the price domain $\text{pdom}(u)$ hold:*

- (i) for any $\mathbf{q} \in \text{pdom}(u)$, $\max_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b \log u(y) - \langle \mathbf{q}, y \rangle < +\infty$;
- (ii) $\{\mathbf{q} \mid \mathbf{q} > \mathbf{0}\} \subseteq \text{pdom}(u)$;
- (iii) for any $\mathbf{q} \geq \mathbf{q}' \geq \mathbf{0}$ such that $\{j : q'_j = 0\} = \{j : q_j = 0\}$, $\mathbf{q} \in \text{pdom}(u)$ implies $\mathbf{q}' \in \text{pdom}(u)$;
- (iv) for any $\mathbf{p} \geq \mathbf{0}$, if $\max_{\langle \mathbf{p}, y \rangle \leq b} \log u(y) < +\infty$, then $\mathbf{p} \in \text{pdom}(u)$.
- (v) for any $M > 0$, if there is a convergent sequence of prices $\mathbf{p}^{(k)} \in \text{pdom}(u)$ with $\max_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b \log u(y) - \langle \mathbf{p}, y \rangle < M$, $k = 1, 2, \dots$ with a limit point $\bar{\mathbf{p}}^{(k)} \rightarrow \bar{\mathbf{p}}$, then $\bar{\mathbf{p}} \in \text{pdom}(u)$.

Proof. We first prove property (i). According to the definition of $\text{pdom}(u)$, there exists $b' > 0$ such that $\max_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b' \log u(y) - \langle \mathbf{q}, y \rangle < +\infty$. For any $b > 0$ and for any $y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$, if $\langle \mathbf{q}, y \rangle \leq b$, then $b \log u(y) - \langle \mathbf{q}, y \rangle < \frac{b}{b'} (b' \log u(y) - \langle \mathbf{q}, y \rangle) + \frac{b-b'}{b'} \langle \mathbf{q}, y \rangle$, which is bounded. Then, the result follows using Lemma 5.1.

Property (ii) holds as, by Lemma 5.1, $\max_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b \log u(y) - \langle \mathbf{q}, y \rangle \leq \max_{\langle \mathbf{q}, x \rangle \leq b} b \log u(x) - \langle \mathbf{q}, x \rangle$, which is bounded as x is bounded: $x_j \leq \frac{b}{q_j} < \infty$.

To prove property (iii), we let $\alpha = \min_{q_j > 0} \{q'_j/q_j\}$. Then, $\max_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} (\alpha b_i) \log u_i(y_i) - \langle \mathbf{q}', y_i \rangle \leq \max_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} (\alpha b_i) \log u_i(y_i) - \alpha \langle \mathbf{q}, y_i \rangle < +\infty$.

Property (iv) holds as, for any y such that $\langle \bar{\mathbf{p}}, y \rangle \leq b$, $b \log u(y) - \langle \bar{\mathbf{p}}, y \rangle \leq b \log u(y) - \langle \mathbf{q}, y \rangle < +\infty$.

Finally, let us prove (v). For a contradiction, assume $\bar{\mathbf{p}} \notin \text{pdom}(u)$, that is, $\sup_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b \log u(y) - \langle \bar{\mathbf{p}}, y \rangle = \infty$. Consider a vector $y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ with $b \log u(y) - \langle \bar{\mathbf{p}}, y \rangle > 2M$. Clearly, $y \neq 0$. Let us select k such that $\|\mathbf{p}^{(k)} - \bar{\mathbf{p}}\| < M/\|y\|$. Then, $b \log u(y) - \langle \mathbf{p}^{(k)}, y \rangle = b \log u(y) - \langle \bar{\mathbf{p}}, y \rangle + \langle \bar{\mathbf{p}} - \mathbf{p}^{(k)}, y \rangle > M$, in contradiction with the assumption on the sequence $\mathbf{p}^{(k)}$. $\square$

We next prove Lemma 2.1.

LEMMA 2.1. *Under Assumption 1, for any $\mathbf{p} \in \text{pdom}(u)$ and any $b > 0$, $\mathcal{GD}^u(\mathbf{p}, b) \neq \emptyset$.*

Proof. Let $M \stackrel{\text{def}}{=} \sup_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b \log u(y) - \langle \mathbf{p}, y \rangle$, which implies the existence of a sequence of allocation $\{y^{(k)}\}_{k=1}^{\infty}$ such that $b \log u(y^{(k)}) - \langle \mathbf{p}, y^{(k)} \rangle$ converges to M . We show the existence of an allocation y^* satisfying $b \log u(y^*) - \langle \mathbf{p}, y^* \rangle = M$.

According to Lemma 5.1, there exists a sequence of $\bar{y}^{(k)}$ such that $b \log u(\bar{y}^{(k)}) - \langle \mathbf{p}, \bar{y}^{(k)} \rangle$ converges to M , with $0 \leq \langle \mathbf{p}, \bar{y}^{(k)} \rangle \leq b$ for all k . Thus, we can select a subsequence $\bar{y}^{(i_k)}$ such that $b \log u(\bar{y}^{(i_k)}) - \langle \mathbf{p}, \bar{y}^{(i_k)} \rangle$ converges to M and $\langle \mathbf{p}, \bar{y}^{(i_k)} \rangle$ converges to b' for some $b' \in [0, b]$. Consequently, $b \log u(x^{(i_k)})$ converges to $M + b'$ as $\langle \mathbf{p}, x^{(i_k)} \rangle$ converges to b' . By Assumption 1, there exists an allocation y^* such that $b \log u(y^*) = M + b'$ and $\langle \mathbf{p}, y^* \rangle \leq b'$. This completes the proof. $\square$

We can use the subgradient Karush–Kuhn–Tucker (KKT) conditions to characterize solutions in the Gale demand correspondence. We refer the reader to Lagrangian duality and KKT conditions of nondifferentiable functions to Chapter 3 in [50].

LEMMA 5.3. *For a utility function $u : \mathbb{R}_{\geq 0}^{\mathcal{M}} \rightarrow \mathbb{R}_{\geq 0}$, $b > 0$ and $\mathbf{q} \in \text{pdom}(u)$, let $y \in \mathcal{GD}^u(\mathbf{q}, b)$. Then, there exists a supergradient $\mathbf{g} \in \partial^\wedge u(y)$ such that*

$$g_j \leq q_j \cdot \frac{u(y)}{b}, \quad \forall j \in \mathcal{M}, \text{ with equality if } y_j > 0,$$

Conversely, if such a $\mathbf{g} \in \partial^\wedge u(y)$ exists then $y \in \mathcal{GD}^u(\mathbf{q}, b)$.

We are ready to prove Lemma 3.1

LEMMA 3.1. *Consider a utility function $u : \mathbb{R}_{\geq 0}^{\mathcal{M}} \rightarrow \mathbb{R}_{\geq 0}$, prices $\mathbf{p}, \mathbf{q} \in \text{pdom}(u)$ such that $\mathbf{q} \leq \mathbf{p}$. Then, $u(y) \geq \frac{1}{2}u(x)$ for any $y \in \mathcal{GD}^u(\mathbf{q}, b)$ and $x \in \mathcal{D}^u(\mathbf{p}, b)$.*

Proof. By Lemma 5.3, there exists a supergradient $\mathbf{g} \in \partial^\wedge u(y)$ such that $g_j \leq q_j u(y)/b$ with equality whenever $y_j > 0$. Therefore,

$$\begin{aligned} u(x) &\leq u(y) + \langle \mathbf{g}, x - y \rangle \leq u(y) + \frac{u(y)}{b} \langle \mathbf{q}, x - y \rangle \\ &\leq u(y) \left(1 + \frac{\langle \mathbf{q}, x \rangle}{b} \right) \leq u(y) \left(1 + \frac{\langle \mathbf{p}, x \rangle}{b} \right) \leq 2u(y). \end{aligned}$$

$\square$

5.1 Nash welfare as approximate competitive equilibrium We now derive Theorem 1.2, showing that Nash welfare maximizing allocations can be seen as approximate competitive equilibria.

THEOREM 5.1. (RESTATEMENT OF THEOREM 1.2) *Consider a Fisher market instance, and let $\mathbf{y}$ be an allocation maximizing Nash welfare along with Gale prices $\mathbf{q}$. Then, $(\mathbf{y}, \mathbf{q})$ forms a 2-demand-approximate competitive equilibrium. Additionally, when all b_i 's are identical, the allocation is 2-envy-free.*

Proof. By Lemma 5.1, all agents satisfy the budget constraints. Additionally, the market clears by the KKT conditions for (EG). For the approximate competitive equilibrium, we only need to show $u_i(y_i) \geq \frac{1}{2} \max_{\langle \mathbf{q}, x_i \rangle \leq b_i} u_i(x_i)$ for all agents.

Note that y_i maximizes $b_i \log u_i(y_i) - \langle \mathbf{q}, y_i \rangle$. By Lemma 5.3, there is a supergradient g_i such that $g_{ij} \leq q_j u_i(y_i)/b_i$ with equality whenever $y_{ij} > 0$. Then, for any $x_i \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ such that $\langle \mathbf{q}, x_i \rangle \leq b_i$,

$$u_i(x_i) \leq u_i(y_i) + \langle g_i, x_i - y_i \rangle \leq u_i(y_i) + \frac{u_i(y_i)}{b_i} \langle \mathbf{q}, x_i - y_i \rangle \leq \left(2 - \frac{\langle \mathbf{q}, y_i \rangle}{b_i}\right) u_i(y_i) \leq 2u_i(y_i).$$

This proves that $(\mathbf{y}, \mathbf{q})$ forms a 2-approximate competitive equilibrium. Additionally, allocation is 2-envy-free follows directly from Lemma 2.2. $\square$

6 Efficiency of Nash welfare for Gale-substitutes

In this section, we prove Theorem 1.3. We show that for Σ -Gale substitutes utility functions, given a competitive equilibrium $(\mathbf{x}, \mathbf{p})$, there are Gale prices $\mathbf{q}^*$ such that $\mathbf{q}^* \leq \mathbf{p}$. The statement then follows by Lemma 3.1. We start by re-stating and proving Lemma 3.2.

LEMMA 3.2. *Let $u(\cdot)$ be a Σ -Gale-substitutes utility and let $x \in \mathcal{D}^u(\mathbf{p}, b)$ for $\mathbf{p} \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ and $b > 0$. Then, $\mathbf{p} \in \text{pdom}(u)$, and there exists $y \in \mathcal{GD}^u(\mathbf{p}, b)$ such that $y \leq x$.*

Proof. According to the KKT conditions, there exists $\lambda \geq 0$ and supergradient $\mathbf{g} \in \partial^+ u(x)$ such that

$$(6.6) \quad g_j \leq \lambda p_j, \quad \forall j \in \mathcal{M}, \text{ with equality if } x_j > 0,$$

and $\langle \mathbf{p}, x \rangle = b$ if $\lambda > 0$.

Let us first consider the case $\lambda = 0$. Then, $g_j \leq 0$ with equality if $x_j > 0$, implying that $x = \arg \max_{x \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} u(x)$. Consequently, $u(\cdot)$ is satiable; therefore, $\mathbf{0} \in \text{pdom}(u)$ and $x \in \mathcal{GD}^u(\mathbf{0}, b)$. Now, (Σ) guarantees the existence of $y \in \mathcal{GD}^u(\mathbf{p}, b)$, $y \leq x$. Clearly, $\text{pdom}(u) = \mathbb{R}_{\geq 0}^{\mathcal{M}}$ in this case.

Next, assume $\lambda > 0$. Let us define

$$\mathbf{q} \stackrel{\text{def}}{=} \frac{\lambda b}{u(x)} \cdot \mathbf{p}.$$

Note that $\mathbf{q} \leq \mathbf{p}$. This follows since $\lambda b = \lambda \langle \mathbf{p}, x \rangle = \langle \mathbf{g}, x \rangle \leq u(x)$, where the final inequality is by concavity and $u(\mathbf{0}) = 0$. Thus,

$$g_j \leq \frac{u(x)}{b} q_j, \quad \forall j \in \mathcal{M}, \text{ with equality if } x_j > 0.$$

According to Lemma 5.3, $x \in \mathcal{GD}^u(\mathbf{q}, b)$.

Note that $\mathcal{GD}^u(\mathbf{q}, b) = \mathcal{GD}^u(\mathbf{p}, u(x)/\lambda)$. Utilizing Definition 2.10(ii) for $x \in \mathcal{GD}^u(\mathbf{p}, u(x)/\lambda)$ and $b \leq u(x)/\lambda$, there exists $y \in \mathcal{GD}^u(\mathbf{p}, b)$ such that $y \leq x$ as required. $\square$

We next show that for any competitive equilibrium $(\mathbf{x}, \mathbf{p})$, and any prices $\mathbf{q} \leq \mathbf{p}$, there exist an aggregate Gale demand such that $\sum_{i \in \mathcal{A}} y_{ij} \leq 1$ whenever $q_j = p_j$.

LEMMA 6.1. *Consider a Fisher market instance such that $u_i(\cdot)$ is Σ -Gale-substitute utilities for all agents. For any competitive equilibrium $(\mathbf{x}, \mathbf{p})$, for any $\mathbf{q} \in \cap_{i \in \mathcal{A}} \text{pdom}(u_i)$ such that $\mathbf{q} \leq \mathbf{p}$, there exists $y_i \in \mathcal{GD}_i(\mathbf{q}, b_i)$ for all $i \in \mathcal{A}$ and $\sum_{i \in \mathcal{A}} y_{ij} \leq 1$ for all $j \in \mathcal{M}$ such that $q_j = p_j$.*

Proof. By Lemma 3.2, there exists $\bar{y}_i \in \mathcal{GD}_i(\mathbf{p}, b_i)$ for every $i \in \mathcal{A}$ such that $\bar{y}_i \leq x_i$. Using Definition 2.10(i), there exists $y_i \in \mathcal{GD}_i(\mathbf{q}, b_i)$ and $y_{ij} \leq \bar{y}_{ij}$ if $q_j = p_j$. Thus, if $q_j = p_j$ then $\sum_{i \in \mathcal{A}} y_{ij} \leq \sum_{i \in \mathcal{A}} \bar{y}_{ij} \leq \sum_{i \in \mathcal{A}} x_{ij} \leq 1$. $\square$

Recall the Lagrangian dual function φ defined in (2.4) on the domain $\text{dom}(\varphi) = \cap_{i \in \mathcal{A}} \text{pdom}(u_i)$.

$$\varphi(\mathbf{q}) \stackrel{\text{def}}{=} \sum_{i \in \mathcal{A}} \max_{y_i \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} (b_i \log u_i(y_i) - \langle \mathbf{q}, y_i \rangle) + \sum_{j \in \mathcal{M}} q_j.$$

The following properties are easy to verify.

LEMMA 6.2. *The following hold for the function $\varphi(\cdot)$.*

- (i) $\varphi(\mathbf{q})$ is convex.
- (ii) Given $\mathbf{q} \in \text{dom}(\varphi)$, let $y_i \in \mathcal{GD}_i(\mathbf{q}, b_i)$ for all $i \in \mathcal{A}$. Then, $\mathbf{1} - \sum_{i \in \mathcal{A}} y_i \in \partial^\vee \varphi(\mathbf{q})$.
- (iii) The optimum value of (2.5), that is, the infimum of $\varphi(\mathbf{q})$ is taken in $\text{dom}(\varphi)$.

Proof. [Proof of Theorem 1.3] Let $(\mathbf{x}, \mathbf{p})$ be a competitive equilibrium solution. Consider the dual (2.5) of the Eisenberg-Gale program (EG), i.e.,

$$\min_{\mathbf{q} \in \text{dom}(\varphi)} \varphi(\mathbf{q}).$$

We show that this has an optimal solution with $\mathbf{q} \leq \mathbf{p}$. The theorem then follows by Lemma 3.1. Let us take any optimal $\mathbf{q}^* \in \text{dom}(\varphi)$; we are done if $\mathbf{q}^* \leq \mathbf{p}$. Otherwise, let $J \stackrel{\text{def}}{=} \{j \in \mathcal{M} : \mathbf{q}_j^* > p_j\}$, and let us define $\bar{\mathbf{q}} \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ as

$$\bar{q}_j \stackrel{\text{def}}{=} \begin{cases} p_j, & \text{if } j \in J, \\ q_j^*, & \text{if } j \notin J. \end{cases}$$

First, assuming $\bar{\mathbf{q}} \in \text{dom}(\varphi)$, we show that $\bar{\mathbf{q}}$ is also optimal to (2.5) as claimed.

Let us pick an allocation $\mathbf{y}$ for the prices $\bar{\mathbf{q}}$ as guaranteed by Lemma 6.1: $\sum_{i \in \mathcal{A}} y_{ij} \leq 1$ for all $j \in \mathcal{M}$ with $\bar{q}_j = p_j$; in particular, for all $j \in J$. According to Lemma 6.2(ii), we get a subgradient $\bar{\mathbf{g}} \in \partial^\vee \varphi(\bar{\mathbf{q}})$ defined as $\bar{\mathbf{g}} \stackrel{\text{def}}{=} \mathbf{1} - \sum_{i \in \mathcal{A}} y_i$; this has the property $\bar{g}_j \geq 0$ for all $j \in J$. We then have

$$\varphi(\mathbf{q}^*) \geq \varphi(\bar{\mathbf{q}}) + \langle \bar{\mathbf{g}}, \mathbf{q}^* - \bar{\mathbf{q}} \rangle = \varphi(\bar{\mathbf{q}}) + \sum_{j \in J} \bar{g}_j (q_j^* - \bar{q}_j) \geq \varphi(\bar{\mathbf{q}}).$$

Here, the first inequality is by the definition of subgradients; the equality uses that $q_j^* = \bar{q}_j$ if $j \notin J$, and the final inequality follows since $\bar{g}_j \geq 0$ and $q_j^* > \bar{q}_j$ for $j \in J$. This shows that $\bar{\mathbf{q}}$ is also an optimal solution to (2.5), completing the proof assuming $\bar{\mathbf{q}} \in \text{dom}(\varphi)$.

It is left to show that $\bar{\mathbf{q}} \in \text{dom}(\varphi)$. Let $K \stackrel{\text{def}}{=} \{j \in \mathcal{M} : \mathbf{q}_j^* = 0 < p_j\}$. Clearly, $K \cap J = \emptyset$. If $K = \emptyset$, then Lemma 5.2(iii) already implies $\bar{\mathbf{q}} \in \text{pdom}(u)$, since $\mathbf{p} \geq \bar{\mathbf{q}}$ and $\text{supp}(\bar{\mathbf{q}}) = \text{supp}(\mathbf{p})$. Otherwise, for $k = 1, 2, \dots$, let us define $\mathbf{q}^{(k)} \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$ as

$$q_j^{(k)} \stackrel{\text{def}}{=} \begin{cases} p_j, & \text{if } j \in J, \\ \frac{p_j}{k}, & \text{if } j \in K, \\ q_j^*, & \text{if } j \notin J \cup K. \end{cases}$$

We have $\mathbf{q}^{(k)} \in \text{pdom}(u)$ again by Lemma 5.2(iii). As in the first part, we take a subgradient $\mathbf{g}^{(k)}$ as guaranteed by Lemma 6.2(ii), with the properties that $\bar{g}_j \geq 0$ for all $j \in J$ and also $\mathbf{g}^{(k)} \leq \mathbf{1}$. We can write

$$\varphi(\mathbf{q}^*) \geq \varphi(\mathbf{q}^{(k)}) + \langle \mathbf{g}^{(k)}, \mathbf{q}^* - \mathbf{q}^{(k)} \rangle = \varphi(\mathbf{q}^{(k)}) + \sum_{j \in J} \bar{g}_j (q_j^* - q_j^{(k)}) - \sum_{j \in K} \bar{g}_j q_j^{(k)} \geq \varphi(\mathbf{q}^{(k)}) - \frac{\|\mathbf{p}\|_\infty}{k}.$$

Here, we use in the first equality that $q_j^* = q_j^{(k)}$ for $j \notin J \cup K$, and $q_j^* = 0$ for $j \in K$. The last inequality uses $\bar{g}_j \leq 1$ and the definition $q_j^{(k)} = p_j/(k+1)$. Thus, Lemma 5.2(v) is applicable with $M = \varphi(\mathbf{q}^*) + \|\mathbf{p}\|_\infty$, using that for each $j \in \mathcal{M}$, $\max_{y_i \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} b_i \log u_i(y_i) - \langle \mathbf{q}^{(k)}, y_i \rangle \leq \varphi(\mathbf{q}^{(k)})$. This shows that $\bar{\mathbf{q}} \in \text{pdom}(u_i)$ for all $i \in \mathcal{M}$, and therefore $\bar{\mathbf{q}} \in \text{dom}(\varphi) = \cap_{i \in \mathcal{M}} \text{pdom}(u_i)$. $\square$

7 Separable utilities are Σ -Gale-substitutes

In this section, we prove Theorem 1.4.

THEOREM 7.1. *Every separable utility function is Σ -Gale-substitutes.*

Recall the following simple monotonicity property of supergradients of univariate concave functions.

LEMMA 7.1. *Let $v : \mathbb{R} \rightarrow \mathbb{R}$ be a univariate concave function, $\alpha, \alpha' \in \text{dom}(f)$, and $g \in \partial^\wedge v(\alpha)$, $g' \in \partial^\wedge v(\alpha')$. Then, $\alpha < \alpha'$ implies $g \geq g'$. Conversely, if $g > g'$, then $\alpha \leq \alpha'$.*

We now adapt Lemma 5.3 to separable utilities.

LEMMA 7.2. *Let $u(x) = \sum_{j \in \mathcal{M}} v_j(x_j)$ be a separable utility function. For any $b > 0$, $\mathbf{q} \in \text{pdom}(u)$, $y \in \mathcal{GD}^u(\mathbf{q}, b)$, we have $u(y) > 0$. Further, $y \in \mathcal{GD}^u(b, \mathbf{q})$ if and only if there exist supergradients $g_j \in \partial^\wedge v_j(y_j)$, $j \in \mathcal{M}$ satisfying*

- (i) $g_j \geq 0$; and
- (ii) $g_j \leq q_j u(y)/b$ with equality whenever $y_j > 0$.

The proof of Theorem 1.4 follows by the next three lemmas, that prove the three properties in Definition 2.10 for separable utilities: property (i) in Lemma 7.3, property (ii) in Lemma 7.4, and property (Σ) in Lemma 7.5.

LEMMA 7.3. *Let $u(x) = \sum_{j \in \mathcal{M}} v_j(x_j)$ be a separable utility function, and let $b > 0$, $\mathbf{q}, \mathbf{q}' \in \text{pdom}(u)$ and $y \in \mathcal{GD}^u(\mathbf{q}, b)$. If $\mathbf{q} \geq \mathbf{q}'$, then there exists a $y' \in \mathcal{GD}^u(\mathbf{q}', b)$ such that $y'_j \leq y_j$ whenever $q'_j = q_j$.*

Proof. Let $J \stackrel{\text{def}}{=} \{j \in \mathcal{M} : q'_j = q_j\}$. Let us select $y' \in \mathcal{GD}^u(\mathbf{q}', b)$ minimizing the potential $\Psi(y') \stackrel{\text{def}}{=} \sum_{j \in J} (y'_j - y_j)^+$. Our goal is to show $\Psi(y') = 0$. Throughout the proof, let $g'_j \in \partial^\wedge v_j(y'_j)$ and $g_j \in \partial^\wedge v_j(y_j)$ as in Lemma 7.2.

First, we show $u(y') \geq u(y)$. For a contradiction, assume $u(y') < u(y)$. Then, according to Lemma 7.2, for every $j \in \mathcal{M}$ with $y_j > 0$ and $q_j > 0$ we have

$$g'_j \leq q'_j u(y')/b < q_j u(y)/b = g_j,$$

while for $j \in \mathcal{M}$ with $q_j = 0$ we have $g'_j = g_j = 0$. Consequently, for each $y_j > 0$, if $q_j > 0$ then Lemma 7.1 implies $y'_j \geq y_j$, and if $q_j = 0$ then $v_j(y_j) = v_j(y'_j)$, both being equal to the maximum value of $v_j(\cdot)$. These imply $u(y') \geq u(y)$, a contradiction.

Using the above, for any $j \in J$ where $y'_j > 0$,

$$(7.7) \quad g'_j = \frac{q'_j}{b} u(y') \geq \frac{q_j}{b} u(y) \geq g_j.$$

Let us first consider the case $u(y') > u(y)$. For all $j \in J$ with $q'_j = q_j > 0$, we get $g'_j > g_j$. By Lemma 7.1, $y'_j \leq y_j$ follows for such goods j . On the other hand, if $q'_j = q_j = 0$, then this implies $g'_j = g_j = 0$. For such indices, if $y'_j > y_j$, then replacing y'_j by y_j results in another bundle $y'' \in \mathcal{GD}^u(\mathbf{q}', b)$ with $\Psi(y'') < \Psi(y')$, a contradiction.

For the rest of the proof, let us assume that $u(y') = u(y)$. We next show $y'_j \geq y_j$ for all $j \in \mathcal{M} \setminus J$, that is, for all goods with $q'_j < q_j$. This is clearly true if $y_j = 0$. Whenever $y_j > 0$ for such an index, we have

$$g'_j \leq \frac{q'_j}{b} u(y') < \frac{q_j}{b} u(y) = g_j,$$

and therefore $y'_j \geq y_j$ by Lemma 7.1. By monotonicity, $v_j(y'_j) \geq v_j(y_j)$ for all $j \in \mathcal{M} \setminus J$. Since $u(y') = u(y)$, we have $\sum_{j \in J} v_j(y'_j) \leq \sum_{j \in J} v_j(y_j)$.

From (7.7), it follows that $g'_j = g_j$ for all $j \in J$ with $y'_j > 0$. This is because the first inequality is now equality, and the second inequality may only be strict if $y_j = 0 < y'_j$. By Lemma 7.1, $g_j \geq g'_j$ in this case, leading to a contradiction.

Assume for a contradiction $\Psi(y') > 0$, and let us select an index $k \in J$ with $y'_k > y_k$. If $v_k(y'_k) = v_k(y_k)$, then we get a contradiction, because changing y'_k to y_k would preserve the optimality conditions in Lemma 7.2, while decreasing the potential $\Psi(y')$. Thus, $v_k(y'_k) > v_k(y_k)$. By $\sum_{j \in J} v_j(y'_j) \leq \sum_{j \in J} v_j(y_j)$, there must be an index $j \in J$ with $v_j(y'_j) < v_j(y_j)$. We can now find suitable $\varepsilon_k, \varepsilon_j > 0$ such that $v_k(y'_k - \varepsilon_k) + v_j(y'_j + \varepsilon_j) = v_k(y'_k) + v_j(y'_j)$, and $y'_j + \varepsilon_j \leq y_j$. This modified solution y'' satisfies the optimality conditions but $\Psi(y'') < \Psi(y')$, again leading to a contradiction. $\square$

LEMMA 7.4. Let $u(x) = \sum_{j \in \mathcal{M}} v_j(x_j)$ be a separable utility function, $b > b' > 0$ and $\mathbf{q} \in \text{pdom}(u)$. Then, for any $y \in \mathcal{GD}^u(\mathbf{q}, b)$, there exists a $y' \in \mathcal{GD}^u(\mathbf{q}, b')$ such that $y' \leq y$.

Proof. We argue similarly as in the previous proof: we pick $y' \in \mathcal{GD}^u(\mathbf{q}, b')$ which minimizes $\Psi(y') \stackrel{\text{def}}{=} \sum_{j \in \mathcal{J}} (y'_j - y_j)^+$, and show that $\Psi(y') = 0$.

First, we show that $u(y)/b \leq u(y')/b'$. For a contradiction, suppose $u(y)/b > u(y')/b'$. If $y_j > 0$, then, by Lemma 7.2

$$g_j = \frac{q_j u(y)}{b} > \frac{q_j u(y')}{b'} \geq g'_j,$$

which implies $v_j(y_j) \leq v_j(y'_j)$ whenever $y_j > 0$ (Lemma 7.1), and therefore, $u(y) \leq u(y')$. Together with $b > b'$, this implies $u(y)/b \leq u(y')/b'$, a contradiction.

Whenever $y'_j > 0$, we now have

$$(7.8) \quad g'_j = \frac{q_j u(y')}{b'} \geq \frac{q_j u(y)}{b} \geq g_j.$$

Assume first $u(y')/b' > u(y)/b$. If $q_j > 0$, then this inequality is strict, and therefore $y'_j \leq y_j$ for all $j \in \mathcal{M}$ follows by Lemma 7.1. If $q_j = 0$, then $g'_j = g_j = 0$. If $y'_j > y_j$ for such a good j , we can decrease y'_j to y_j and thereby get an optimal allocation with strictly smaller $\Psi(y')$ value, leading to a contradiction. This completes the proof for $u(y')/b' > u(y)/b$.

For the rest, we may assume $u(y')/b' = u(y)/b$. Now, $g'_j = g_j$ must hold whenever $y'_j > 0$. This is because the first inequality in (7.8) is equality, and if the second is strict, then $y_j = 0 < y'_j$ would give a contradiction to $g'_j > g_j$.

Assume for a contradiction $\Psi(y') > 0$, and let us pick $k \in \mathcal{M}$ with $y'_k > y_k$. The equality $u(y')/b' = u(y)/b$ and $b' < b$ gives $u(y') < u(y)$. We can derive a contradiction by decreasing y'_k and potentially increasing an y'_j similarly as in the proof of Lemma 7.3. $\square$

LEMMA 7.5. Let $u(x) = \sum_{j \in \mathcal{M}} v_j(x_j)$ be a separable utility function such that $\mathbf{0} \in \text{pdom}(u)$. Then, for any $\mathbf{q} \geq \mathbf{0}$, $b > 0$ and any $y \in \mathcal{GD}^u(\mathbf{0}, b)$, there exists a $y' \in \mathcal{GD}^u(\mathbf{q}, b)$ such that $y' \leq y$.

Proof. $\mathbf{0} \in \text{pdom}(u)$ implies $\sup_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} u(y) < \infty$. Therefore, $\text{pdom}(u) = \mathbb{R}_{\geq 0}^{\mathcal{M}}$; in particular, $\mathbf{q} \in \text{pdom}(u)$. Let us pick $y' \in \mathcal{GD}^u(\mathbf{q}, b)$ such that $\sum_{j \in \mathcal{M}} y'_j$ is minimal.

Note that $u(y) = \sup_{y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}} u(y)$. In particular, for any $\alpha \geq y_j$, $v_j(\alpha) = v_j(y_j)$. For a contradiction, assume there exists a $k \in \mathcal{M}$ with $y_k < y'_k$. Then, reducing y'_k to y_k may only increase $b \log u(y') - \langle \mathbf{q}, y' \rangle$, while decreasing $\sum_{j \in \mathcal{M}} y'_j$. This contradicts the choice of y' . $\square$

8 Discussion

In this paper, we compared competitive equilibrium and Nash welfare maximizing solutions in the Fisher market setting. We identified Gale-substitutes as a broad family of utility functions where the Nash welfare uniformly guarantees every agent at least half of their maximum utility at any competitive equilibrium. We also introduced generalized network utilities. This is a very general class with a rich combinatorial structure and may be a relevant example in various market and fair division settings.

Gale-substitutes may be a class of concave functions worth investigating on its own. In the context of indivisible goods, the analogous class of gross substitute utilities, introduced by Keslo and Crawford [41], turned out to be the same as the important class of M^{h} -concave utilities in discrete convex analysis [29]. In particular, one may wonder whether the Gale-substitutes property implies *submodularity*, that is, $u(x) + u(y) \geq u(x \vee y) + u(x \wedge y)$ for any $x, y \in \mathbb{R}_{\geq 0}^{\mathcal{M}}$, where $x \vee y$ and $x \wedge y$ denote the pointwise maximum and minimum of the two vectors, respectively. This property trivially holds for separable utilities and can also be shown for generalized network utilities.

We note that one can construct an example of a concave quadratic function that is submodular but not Gale-substitutes. On the other hand, it is unclear whether Theorem 1.6 would remain true for submodular utilities instead of Gale-substitutes. Another natural question related to the well-studied notion of *weak gross substitutes* (WGS) utilities: Gale-substitutes includes utility functions that are not WGS, such as concave separable; but it is not clear whether all WGS utilities are also Gale-substitutes.

Gale-substitutes utilities may also have potential algorithmic applications. The WGS property leads to natural tâtonnement and auction mechanisms for finding competitive equilibria, see, e.g., [3, 6, 7, 16, 17, 30]. The Gale-substitutes property may be useful to design simple such mechanisms and combinatorial algorithms for solving the convex program (EG). For the linear case, Devanur et al. [22] gave a primal-dual algorithm that was followed by a series of works on combinatorial algorithms for various market settings; see, e.g., [11, 20, 23, 35].

A Examples

In this section, we present multiple examples mentioned in the paper.

A.1 Connectivity of equilibria with bounded linear utilities We first investigate the (dis)connectivity of competitive equilibria with bounded linear utilities. We begin by demonstrating that the thrifty competitive equilibria with bounded linear utilities might not be connected.

EXAMPLE A.1. (DISCONNECTIVITY OF THRIFTY COMPETITIVE EQUILIBRIUM) There are two agents and two goods. Each agent has a unit budget. The first agent's utility is $u_1(x_1) = 1.3 \min\{x_{11}, 0.8\} + 0.45x_{12}$ and the second agent's utility is $u_2(x_2) = 0.01 \min\{x_{21}, 0.3\} + 2 \min\{x_{22}, 0.8\}$.

First, we show there are at least two equilibria. The first equilibrium is with $\mathbf{p} = (1, 1)$. Under this price, $x_{11} = 0.8$, $x_{12} = 0.2$, $x_{21} = 0.2$, $x_{22} = 0.8$. Another competitive equilibrium is with $\mathbf{p} = (1.3, 0.45)$. Under this price, $x_{11} = 0.7$, $x_{12} = 0.2$, $x_{21} = 0.3$, $x_{22} = 0.8$.

Now, we show the second competitive equilibrium is isolated under the thrifty assumption.

First, we show the price is isolated. Suppose there is a nearby price $\mathbf{p}' = (1.3 + \epsilon_1, 0.45 + \epsilon_2)$. For this price, the thrifty demand of the second agent remains the same: $x_{21} = 0.3$, $x_{22} = 0.8$. To maintain equilibrium, the first agent should receive the same allocation. As $1.3 \times 0.7 + 0.45 \times 0.2 = 1$, ϵ_1 and ϵ_2 can not be both positive or negative. However, if one ϵ is negative, the first agent will prefer this item more at $\mathbf{p}'$. This implies the allocation of the other item of the first agent is 0. This leads to a contradiction.

Second, we show the allocation is isolated. Suppose there is a nearby equilibrium allocation of the second agent $x'_2 = (x_{21} - \epsilon_1, x_{22} - \epsilon_2)$ with at least one of ϵ_1 and ϵ_2 is strictly positive. In this case, both agents should spend all their budgets as they are non-satiable. Therefore, the equilibrium prices satisfy $p'_1 + p'_2 = 2$. For the first agent, the allocation is $x'_{11} = 0.7 + \epsilon_1$, $x'_{12} = 0.2 + \epsilon_2$. Note that both ϵ_1 and ϵ_2 are very tiny. This implies $p'_1 : p'_2 = 1.3 : 0.45$. Therefore, $p'_1 = \frac{2}{1.3+0.45} \times 1.3$ and $p'_2 = \frac{2}{1.3+0.45} \times 0.45$. However, the first agent cannot afford these allocations under these prices.

The next example shows that competitive equilibria are not connected without the thrifty assumption.

EXAMPLE A.2. (DISCONNECTIVITY OF COMPETITIVE EQUILIBRIA) There are three agents and three goods. Each agent has a unit budget. The first agent's utility is $u_1(x_1) = 1.3 \min\{x_{11}, 0.8\} + 0.45x_{12}$, the second agent's utility is $u_2(x_2) = 0.01 \min\{x_{21}, 0.3\} + 2 \min\{x_{22}, 0.8\} + 10^{-10}x_{23}$, and the third agent's utility is $u_3(x_3) = x_{33}$.

First, we show there are at least two equilibrium. The first equilibrium is with $\mathbf{p} = (1, 1, 1)$, and the allocation is $x_1 = (0.8, 0.2, 0)$, $x_2 = (0.2, 0.8, 0)$, and $x_3 = (0, 0, 1)$. The second equilibrium is with $\mathbf{p} = (1.3, 0.45, 1.25)$ and the allocation is $x_1 = (0.7, 0.2, 0)$, $x_2 = (0.3, 0.8, 0.2)$, and $x_3 = (0, 0, 0.8)$.

In order to show our results, we first state the following properties: for any equilibrium,

1. $p_3 > 0$ as the third agent is only interested in the third good;
2. $x_{13} = 0$ as the first agent shows no interest in the third good and its price is strictly positive;
3. $p_1 > 0$ and $p_2 > 0$ as, otherwise, these goods will be over-demanded;
4. $x_{31} = x_{32} = 0$ as the third agent shows no interest in the first two goods and their prices are strictly positive;
5. $p_1 + p_2 + p_3 = 3$ as all agents are non-satiated.

Now, we show the second equilibrium is isolated.

First, we demonstrate that the price is isolated. Suppose there is a nearby equilibrium price $\mathbf{p}' = (1.3 + \epsilon_1, 0.45 + \epsilon_2, 1.25 + \epsilon_3)$. For this price, the allocation of the second agent on the first two items is the same. Therefore, the first agent should receive the same allocation. As $1.3 \times 0.7 + 0.45 \times 0.2 = 1$, ϵ_1 and ϵ_2 can not be both positive or negative. Similar to the previous example, if one of ϵ_1 and ϵ_2 is negative, then the first agent will prefer this item more at this nearby price. This implies the allocation of the other item of the first agent is 0. This leads to the conclusion $\epsilon_1 = \epsilon_2 = \epsilon_3 = 0$.

Second, we show the allocation is isolated. Suppose there is a nearby equilibrium allocation $\mathbf{x}'$ with the equilibrium $\mathbf{p}'$. As $x_{11} < 0.8$, $p'_1 : p'_2 = 1.3 : 0.45$.¹ Additionally, $p'_1 \times x'_{11} + p'_2 \times x'_{12} = 1$. This implies p'_1 is very close to p_1 and p'_2 is

¹Otherwise, the allocation of the first agent will be changed significantly.

very close to p_2 : if $\|x' - x\|_\infty \leq \epsilon$, then $|p'_1 - p_1| \leq O(\epsilon)$ and $|p'_2 - p_2| \leq O(\epsilon)$.² This also implies $|p'_3 - p_3| \leq O(\epsilon)$ as $p'_1 + p'_2 + p'_3 = 3$. Since price is isolated, $p'_1 = p_1$, $p'_2 = p_2$, and $p'_3 = p_3$. Under this price, x is the only allocation.

A.2 Non-monotonicity of Gale Demands The next example shows the somewhat counterintuitive property that increasing the prices may lead to an increase in the Gale utility.

EXAMPLE A.3. (NON-MONOTONICITY OF GALE DEMANDS) There are two goods and a single agent i with $b_i = 1$. The agent's utility function is $u_i(x_i) = \min\{x_{i1} + x_{i2}, 2 + 0.1 \cdot x_{i1} + 0.2 \cdot x_{i2}\}$. When $\mathbf{q} = \left(\frac{0.1}{2/0.9+0.2 \cdot 0.01}, \frac{0.2}{2/0.9+0.2 \cdot 0.01}\right)$, the associated Gale demand is $\mathcal{GD}_i(b_i, \mathbf{q}) = \{(2/0.9, 0.01)\}$ and the utility is $u_i(\mathcal{GD}_i(b_i, \mathbf{q})) = 2/0.9 + 0.2 \cdot 0.01$. When $\mathbf{q}' = \left(\frac{1}{0.01+2/0.85}, \frac{0.2}{0.01+2/0.85}\right)$, the Gale demand becomes $\mathcal{GD}_i(b_i, \mathbf{q}') = \{(0.01, 2/0.85)\}$ and $u_i(\mathcal{GD}_i(b_i, \mathbf{q}')) = 2/0.85 + 0.01$. Clearly, $\mathbf{q} < \mathbf{q}'$, yet $u_i(\mathcal{GD}_i(b_i, \mathbf{q})) < u_i(\mathcal{GD}_i(b_i, \mathbf{q}'))$.

A.3 Tight Example on Price of Anarchy The next example shows that the bound in Theorem 1.1 is tight.

EXAMPLE A.4. (TIGHT EXAMPLE ON PRICE OF ANARCHY) We consider the case with two agents and two goods. The first agent has a budget of 1, and desires the first good: $u_1(x_1) = x_{11}$. The second agent has a budget of $e - 1$ and has a utility function $u_2(x_2) = \epsilon x_{21} + \min\{x_{22}, 1\}$, where $\epsilon > 0$ is sufficiently small. There is a competitive equilibrium with prices $p_1 = e$ and $p_2 = 0$, and allocations $x_1 = (\frac{1}{e}, 0)$ and $x_2 = (\frac{e-1}{e}, 1)$. The Nash welfare objective for this allocation is $(\frac{1}{e}(1 + \epsilon \cdot \frac{e-1}{e})^{e-1})^{\frac{1}{e}}$. In the allocation maximizing Nash welfare, we have $y_1 = (1, 0)$ and $y_2 = (0, 1)$, where the Nash welfare objective is 1. When $\epsilon \rightarrow 0$, the difference of these two scenarios converges to $(\frac{1}{e})^{\frac{1}{e}}$.

A.4 Utility Difference in Different Equilibria The following example shows agents might receive significantly different utilities in different equilibria.

EXAMPLE A.5. (UTILITY DIFFERENCE IN DIFFERENT EQUILIBRIA) Consider a scenario involving n agents and n goods, where each agent has a unit budget. For $1 \leq i \leq n - 1$, agent i 's utility is defined as $u_i(x_i) = \min\{x_{ii}, 1\} + \epsilon x_{in}$, while agent n 's utility is given by $u_n(x_n) = x_{nn}$. There exist at least two competitive equilibria. In the first competitive equilibrium, all goods are priced at 1, and each good i is allocated to agent i for $1 \leq i \leq n$, resulting in a utility of 1 for agent n . However, in the second competitive equilibrium, prices are set to 0 for the first $n - 1$ goods and the price of the n -th good is n . Consequently, goods 1 to $n - 1$ are allocated to agents 1 to $n - 1$ respectively, while the n -th good is uniformly allocated to all agents. In this equilibrium, agent n can only attain a utility of $\approx \frac{1}{n}$.

A.5 Tight Example on Demand-Approximate CE The following example shows Theorem 1.2 is tight.

EXAMPLE A.6. (TIGHT EXAMPLE ON DEMAND-APPROXIMATE CE) Consider the case with two agents and one good. Both agents have unit budgets. The first agent's utility on this good is linear: $u_1(x_1) = x_1$. The second agent's utility is defined as follows: $u_2(x_2) = \alpha x_2$ for $x_2 \leq \frac{1}{\alpha}$ and $u_2(x_2) = 1 + \beta(x_2 - \frac{1}{\alpha})$ for $x_2 > \frac{1}{\alpha}$, where α is a large number and $\beta \leq \alpha$. When $\beta = \frac{1-\frac{1}{\alpha}}{1-\frac{2}{\alpha}}$, the Nash welfare maximizing allocation is $x_1 = 1 - \frac{1}{\alpha}$ and $x_2 = \frac{1}{\alpha}$, with the dual variable q equal to $\frac{1}{x_1} = \frac{1}{1-\frac{1}{\alpha}}$. Under this price, agent 2 can obtain $x'_2 = 1 - \frac{1}{\alpha}$ with his/her budget, resulting in a utility of $1 + \beta(x'_2 - \frac{1}{\alpha}) = 2 - \frac{1}{\alpha}$. In contrast, in the Nash welfare maximizing allocation, agent 2 only receives a utility of 1.

A.6 (In)-approximate Nash Welfare utility by Equilibrium utility In this paper, we show for Σ -Gale-substitutes utilities, the utility of any agent at the Nash welfare maximizing allocation is at least half the maximum utility they may get at any CE. It's natural to ask, inversely, if the utility agent can attain at any CE can approximate the utility they receive at Nash welfare maximizing allocation. We show a constant approximation is not achievable by providing an example in which the CE is unique and there exists an agent whose utility at this CE is much worse than his/her utility at Nash welfare maximizing allocation.

²Note that $p'_1 : p'_2 = 1.3 : 0.45 = p_1 : p_2$. By budget constraint, $\frac{1.3}{0.45}p'_2x'_{11} + p'_2x'_{12} = \frac{1.3}{0.45}p_2x_{11} + p_2x_{12}$. Therefore, $p'_2 - p_2 = \frac{p'_2(\frac{1.3}{0.45}(x_{11}-x'_{11})+(x_{12}-x'_{12}))}{\frac{1.3}{0.45}x_{11}+x_{12}}$. We have a similar result for $p'_1 - p_1$.

EXAMPLE A.7. Consider a scenario with $n + 2$ agents and $n + 1$ goods. Set $\epsilon < 1$. For $1 \leq i \leq n$, let $u_i(x_i) = \min\{x_{ii}, 0.5\} + \epsilon x_{i(n+1)}$. Additionally, set $u_{n+1}(x_{n+1}) = x_{(n+1)(n+1)}$ and $u_{n+2}(x_{n+2}) = \sum_{j=1}^n x_{ij}$. We first demonstrate the unique competitive equilibrium is achieved when $p_j = \frac{2}{n}$ for $1 \leq j \leq n$ and $p_{n+1} = n$. In this equilibrium, $x_{ii} = 0.5$ and $x_{i(n+1)} = \frac{n-1}{n^2}$ for $1 \leq i \leq n$; $x_{(n+1)(n+1)} = \frac{1}{n}$; and $x_{(n+2)j} = 0.5$ for $1 \leq j \leq n$.

We now show this is the only competitive equilibrium. First, for all $j : 1 \leq j \leq n, p_j > 0$. Otherwise, agent $n+2$ would receive infinite goods. Given $p_j > 0$ for any $1 \leq j \leq n, x_{(n+2)j} \geq 0.5$. Consequently, the first n goods share the same prices and $0.5p_j n \leq \sum_j x_{(n+2)j} p_j = 1$ which implies $p_j \leq \frac{2}{n}$. If $p_j < \frac{2}{n}$ for $1 \leq j \leq n$, then $p_{n+1} = \sum_i b_i - \sum_j p_j > n$. In this case, $x_{ii} = 0.5$ for $1 \leq i \leq n$ and $0.5p_j n = 1$ for $1 \leq j \leq n$, which implies the result.

Therefore, agent $n + 1$ receives utility $\frac{1}{n}$ in the unique competitive equilibrium. However, for a small enough ϵ , this agent receives 1 in the maximal Nash welfare solution.

A.7 (In)-approximation for general utilities Our next example demonstrates if the utility function is not Gale substitutes, then agents' competitive equilibrium utilities can be much higher than the Nash welfare maximizing solution.

EXAMPLE A.8. There are three types of agents (A, B, C): 1 type-A agent; n^2 type-B agents; n type-C agents; and four types of goods (O, P, R, S): 1 type-O good; n type-P goods; n^2 type-R goods; n type-S goods. In total, there are $N = n^2 + n + 1$ agents and $M = n^2 + 2n + 1$ goods.

The type-A agent is interested in the type-O good and type-P goods. Type-A agent has a *bounded* linear utility with equal preferences over these type-O and type-P goods: $v_{ij} = 1$ for $i = A$ and $j \in O \cup P$. This agent has an upper bound of 1 on the utility of each good: $u_A(x_A) = \sum_{j \in O \cup P} \min\{x_{Aj}, 1\}$.

For type-B agents, let's label each of these agents by (s, t) for $s \in [1, n]$ and $t \in [1, n]$ and also label each of type-R goods by (s, t) for $s \in [1, n]$ and $t \in [1, n]$. Agent (s, t) is interested in the corresponding (s, t) good in R and the s -th good in P. The utility function of agent $i = (s, t)$ is $\min\{x_{iP_s} + \frac{n}{n+1}, x_{iR_{s,t}}\}$, where x_{iP_s} is agent i 's allocation of P_s and $x_{iR_{s,t}}$ is agent i 's allocation of $R_{s,t}$.

For type-C agents, the s -th type-C agent is only interested in the s -th type-S good and all type-R good with label $(s, \cdot)$. This agent also has a bounded linear utility with equal preference ($v_{ij} = 1$) over these goods and an upper bound, $l_{ij} = 1$, on the utility of each good.

We first consider the maximal Nash welfare solution. Here, k is a parameter, which will be specified later.

The type-A agent receives type-O good and $k + \frac{1}{n+1}$ portion of type-P goods, his/her utility will be $1 + nk + \frac{n}{n+1}$.

Each type-B agent obtains $1 - \frac{k}{n}$ portion of corresponding type-R good and $\frac{1}{n+1} - \frac{k}{n}$ portion of corresponding type-P good, yielding a utility of $1 - \frac{k}{n}$.

The type-C agents receive the remaining goods, resulting in a utility of $1 + k$.

The goal is to maximize $(1 + nk + \frac{n}{n+1})(1 - \frac{k}{n})^{n^2} (1 + k)^n$. By calculation, the optimal k is found to be no larger than $n^{-\frac{1}{2}}$.³ Therefore, the type-A agent receives at most $1 + \sqrt{n} + \frac{n}{n+1}$ utility in the maximal Nash welfare solution.

On the other hand, let's consider the (thrifty) competitive equilibrium where all agents have unit budgets.

The prices of type-O, P, and S goods are 0. and prices of type-R goods are $\frac{n+1}{n}$. In this case, the type-A agent will get the type-O good and type-P goods, resulting in a utility of $n + 1$. Each type-B agent receives $\frac{n}{n+1}$ portion of the corresponding type-R good, leading to a utility of $\frac{n}{n+1}$. The type-C agent acquires 1 portion of the corresponding type-S good and the rest of type-R goods, resulting in a utility of $1 + \frac{n}{n+1}$.

Comparing the type-A agent's utility in the two scenarios, the type-A agent achieves at most $1 + \sqrt{n} + \frac{n}{n+1}$ utility in the maximal Nash welfare solution, whereas the agent attains $n + 1$ utility in the (thrifty) competitive equilibrium.

A.8 Tight Example of Theorem 1.3 The following example shows the bound in Theorem 1.3 is tight.

EXAMPLE A.9. Consider the case that there are n agents and n goods. All agents have bounded linear utilities. Additionally, all agents have the same preferences ($v_{ij} = 1$) on all goods except for the first good. For the first good, only the first agent likes the first good ($v_{11} = 0$) and all other agents don't like this good $v_{i1} = 0$ for $i \neq 1$. Additionally, we assume the upper bound on the utility of each good of each agent is 1, $l_{ij} = 1$ for all i and j .

³First, we take the log and calculate the gradient, which is $\frac{n}{1+nk+\frac{n}{n+1}} + \frac{n}{1+k} - \frac{n}{1-\frac{k}{n}}$. Note that this is a decreasing function with k . So, in order to show the optimal k is no larger than $n^{-\frac{1}{2}}$, we only need to show when $k = n^{-\frac{1}{2}}$, $\frac{n}{1+nk+\frac{n}{n+1}} + \frac{n}{1+k} \leq \frac{n}{1-\frac{k}{n}}$. This is true as $\frac{n}{1-\frac{k}{n}} - \frac{n}{1+k} = \frac{(n+1)k}{1+k-\frac{k}{n}-\frac{k^2}{n}} \geq nk = \frac{1}{k} \geq \frac{n}{1+nk+\frac{n}{n+1}}$.

In this case, for the maximal Nash welfare solution, the utilities of all agents equal 1. However, there exists a competitive equilibrium, at which the price of the first good equals 0 and the prices for all other goods equal $\frac{n}{n-1}$. Then, the first agent can get the first good and $\frac{n-1}{n}$ fraction of the second good, and achieve a utility of $1 + \frac{n-1}{n}$ which tends to 2 in the case $n \rightarrow \infty$.

A.9 Example with Leontief-free utility functions Not all Leontief-free utility functions satisfy property (Σ) . The subsequent example illustrates a scenario where an agent can attain significantly more utility in competitive equilibrium compared to the Nash welfare maximizing allocation when agents have Leontief-free utilities.

EXAMPLE A.10. Consider the scenario with n goods and $n+1$ agents. Agent $i = 1, 2, \dots, n$ only likes good i , has budget $b_i = 1$ and a linear utility function $u_i(x_i) = x_{ii}$. Agent $n+1$ has a substantially larger budget $b_{n+1} = B$, and a utility function $\min\{\sum_{i=1}^k x_{(n+1)i}, (n-1)(1-\epsilon)\}$ with a small $\epsilon > 0$.

There is a competitive equilibrium where the prices are $p_1 = 1$ and $p_2 = p_3 = \dots = p_n = \frac{1}{\epsilon}$. Agent 1 obtains $x_{11} = 1$, the other agents get $x_{ii} = \epsilon$ for $i = 2, \dots, n$, and agent $n+1$ gets $x_{(n+1)1} = 0$ and $x_{(n+1)j} = 1 - \epsilon$ for $j = 2, \dots, n$. The competitive equilibrium holds if $B > \frac{(n-1)(1-\epsilon)}{\epsilon}$.

On the other hand, in the Nash welfare maximizing allocation, when B is big enough, agent $n+1$ receives (approximately) $\frac{(1-\epsilon)(n-1)}{n}$ of each good, with the remaining goods shared among the remaining agents. Agent 1 receives a utility of $\epsilon + \frac{1-\epsilon}{n}$, which is significantly less than the utility received in the competitive equilibrium.

In this case, agent $n+1$'s utility doesn't satisfy property (Σ) . When $\mathbf{p} = \mathbf{0}$, $x_{n+1} = (0, 1 - \epsilon, 1 - \epsilon, \dots, 1 - \epsilon) \in \mathcal{GD}_{n+1}(\mathbf{p}, B)$. However, when $\mathbf{p} = (1, \frac{1}{\epsilon}, \frac{1}{\epsilon}, \dots, \frac{1}{\epsilon})$, for any $x_{n+1} \in \mathcal{GD}_{n+1}(\mathbf{p}, B)$, $x_{(n+1)1} > 0$, which violates property (Σ) .

References

- [1] R. K. Ahuja, T. L. Magnanti, and J. B. Orlin. *Network Flows: Theory, Algorithms, and Applications*. Prentice-Hall, Inc., feb 1993.
- [2] N. Anari, T. Mai, S. O. Gharan, and V. V. Vazirani. Nash social welfare for indivisible items under separable, piecewise-linear concave utilities. In *Proceedings of the 29th Symposium Discrete Algorithms (SODA)*, pages 2274–2290, 2018.
- [3] K. J. Arrow, H. D. Block, and L. Hurwicz. On the stability of the competitive equilibrium, II. *Econometrica: Journal of the Econometric Society*, pages 82–109, 1959.
- [4] K. J. Arrow and G. Debreu. Existence of an equilibrium for a competitive economy. *Econometrica: Journal of the Econometric Society*, pages 265–290, 1954.
- [5] K. J. Arrow and F. H. Hahn. *Competitive Equilibrium Analysis*. Holden-Day, 1971.
- [6] K. J. Arrow and L. Hurwicz. Competitive stability under weak gross substitutability: The “Euclidean distance” approach. *International Economic Review*, 1(1):38–49, 1960.
- [7] X. Bei, J. Garg, and M. Hoefer. Ascending-price algorithms for unknown markets. *ACM Transactions on Algorithms (TALG)*, 15(3):1–33, 2019.
- [8] X. Bei, J. Garg, M. Hoefer, and K. Mehlhorn. Earning and utility limits in Fisher markets. *ACM Transactions on Economics and Computation (TEAC)*, 7(2):1–35, 2019.
- [9] S. Brânzei, V. Gkatzelis, and R. Mehta. Nash social welfare approximation for strategic agents. *Oper. Res.*, 70(1):402–415, 2022.
- [10] I. Caragiannis, D. Kurokawa, H. Moulin, A. D. Procaccia, N. Shah, and J. Wang. The unreasonable fairness of maximum Nash welfare. *ACM Trans. Economics and Comput.*, 7(3):12:1–12:32, 2019.
- [11] B. R. Chaudhury, J. Garg, P. McGlaughlin, and R. Mehta. Competitive equilibrium with chores: Combinatorial algorithm and hardness. In *Proceedings of the 23rd Conference on Economics and Computation (EC)*, pages 1106–1107, 2022.

- [12] X. Chen, D. Dai, Y. Du, and S. Teng. Settling the complexity of Arrow-Debreu equilibria in markets with additively separable utilities. In *Proceedings of the 50th Symposium Foundations of Computer Science (FOCS)*, pages 273–282, 2009.
- [13] X. Chen, D. Paparas, and M. Yannakakis. The complexity of non-monotone markets. In *Proceedings of the 45th Symposium Theory of Computing (STOC)*, pages 181–190, 2013.
- [14] X. Chen and S. Teng. Spending is not easier than trading: On the computational equivalence of Fisher and Arrow-Debreu equilibria. In *Proceedings of the 20th International Symposium Algorithms and Computation (ISAAC)*, pages 647–656, 2009.
- [15] B. Codenotti, B. McCune, S. Penumatcha, and K. R. Varadarajan. Market equilibrium for CES exchange economies: Existence, multiplicity, and computation. In *Proceedings of the 25th Conference on Foundations of Software Tech. and Theoret. Comp. Sci. (FSTTCS)*, pages 505–516, 2005.
- [16] B. Codenotti, B. McCune, and K. Varadarajan. Market equilibrium via the excess demand function. In *Proceedings of the thirty-seventh annual ACM symposium on Theory of computing*, pages 74–83, 2005.
- [17] B. Codenotti, S. V. Pemmaraju, and K. R. Varadarajan. On the polynomial time computation of equilibria for certain exchange economies. In *SODA*, volume 5, pages 72–81, 2005.
- [18] B. Codenotti, A. Saberi, K. Varadarajan, and Y. Ye. Leontief economies encode two-player zero-sum games. In *Proceedings of the 17th Symposium Discrete Algorithms (SODA)*, pages 659–667, 2006.
- [19] B. Codenotti and K. Varadarajan. Computation of market equilibria by convex programming. In N. Nisan, É. Tardos, T. Roughgarden, and V. Vazirani, editors, *Algorithmic Game Theory*, chapter 8. Cambridge University Press, 2007.
- [20] R. Cole and V. Gkatzelis. Approximating the Nash social welfare with indivisible items. In *Proceedings of the 47th Symposium Theory of Computing (STOC)*, pages 371–380, 2015.
- [21] A. Deligkas, J. Fearnley, A. Hollender, and T. Melissourgos. Constant inapproximability for Fisher markets. In *Proceedings of the 25th Conference on Economics and Computation (EC)*, 2024.
- [22] N. Devanur, C. Papadimitriou, A. Saberi, and V. Vazirani. Market equilibrium via a primal–dual algorithm for a convex program. *J. ACM*, 55(5), 2008.
- [23] R. Duan and K. Mehlhorn. A combinatorial polynomial algorithm for the linear Arrow-Debreu market. *Inf. Comput.*, 243:112–132, 2015.
- [24] P. Dubey. Inefficiency of Nash equilibria. *Mathematics of Operations Research*, 11(1):1–8, 1986.
- [25] E. Eisenberg. Aggregation of utility functions. *Management Sci.*, 7(4):337–350, 1961.
- [26] E. Eisenberg and D. Gale. Consensus of subjective probabilities: The pari-mutuel method. *The Annals of Mathematical Statistics*, 30(1):165–168, 1959.
- [27] M. Feldman, S. Mauras, and T. Ponitka. On optimal tradeoffs between EFX and Nash welfare. In *Proceedings of the 38th Conference on Artificial Intelligence (AAAI)*, 2024.
- [28] A. Filos-Ratsikas, K. A. Hansen, K. Høgh, and A. Hollender. PPAD-membership for problems with exact rational solutions: A general approach via convex optimization. In *Proceedings of the 56th Symposium Theory of Computing (STOC)*, 2024.
- [29] S. Fujishige and Z. Yang. A note on Kelso and Crawford’s gross substitutes condition. *Mathematics of Operations Research*, 28(3):463–469, 2003.
- [30] J. Garg, E. Huseinović, and L. A. Végh. An auction algorithm for market equilibrium with weak gross substitute demands. *ACM Transactions on Economics and Computation*, 11(3-4):1–24, 2023.

- [31] J. Garg, R. Mehta, M. Sohoni, and V. V. Vazirani. A complementary pivot algorithm for market equilibrium under separable, piecewise-linear concave utilities. *SIAM J. Comput.*, 44(6):1820–1847, 2015.
- [32] J. Garg, R. Mehta, and V. V. Vazirani. Dichotomies in equilibrium computation, and complementary pivot algorithms for a new class of non-separable utility functions. In *Proceedings of the forty-sixth annual ACM symposium on Theory of computing (STOC)*, pages 525–534, 2014.
- [33] J. Garg, Y. Tao, and L. A. Végh. Approximating equilibrium under constrained piecewise linear concave utilities with applications to matching markets. In *Proceedings of the Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 2269–2284, 2022.
- [34] J. Garg, T. Tröbst, and V. V. Vazirani. A nash-bargaining-based mechanism for one-sided matching markets and dichotomous utilities. In *Proceedings of the 2023 International Conference on Autonomous Agents and Multiagent Systems, AAMAS*, pages 2721–2723, 2023.
- [35] J. Garg and L. A. Végh. A strongly polynomial algorithm for linear exchange markets. In *Proceedings of the 51st Symposium Theory of Computing (STOC)*, 2019.
- [36] A. V. Goldberg, S. A. Plotkin, and É. Tardos. Combinatorial algorithms for the generalized circulation problem. *Mathematics of Operations Research*, 16(2):351–381, 1991.
- [37] M. Goyal, S. Sarvasarkar, and A. Goel. A mechanism for participatory budgeting with funding constraints and project interactions. In *Proceedings of the 19th Conference on Web and Internet Economics (WINE)*, pages 329–347, 2023.
- [38] F. Gul, W. Pesendorfer, and M. Zhang. The efficient allocation of indivisible goods. *working paper*, 2022.
- [39] M. Hosseini and V. V. Vazirani. Nash-bargaining-based models for matching markets: one-sided and two-sided; Fisher and Arrow–Debreu. In M. Braverman, editor, *13th Innovations in Theoretical Computer Science Conference, ITCS*, pages 86:1–86:20, 2022.
- [40] A. Hylland and R. Zeckhauser. The efficient allocation of individuals to positions. *Journal of Political economy*, 87(2):293–314, 1979.
- [41] A. S. J. Kelso and V. P. Crawford. Job matching, coalition formation, and gross substitutes. *Econometrica: Journal of the Econometric Society*, pages 1483–1504, 1982.
- [42] E. Koutsoupias and C. Papadimitriou. Worst-case equilibria. In *Proceedings of the 16th Symposium Theoret. Aspects of Computer Science (STACS)*, pages 404–413. Springer, 1999.
- [43] A. Mas-Colell. Equilibrium theory with possibly satiated preferences. *Equilibrium and dynamics: Essays in honour of David Gale*, pages 201–213, 1992.
- [44] L. McKenzie. On equilibrium in Graham’s model of world trade and other competitive systems. *Econometrica: Journal of the Econometric Society*, pages 147–161, 1954.
- [45] H. Moulin. *Fair Division and Collective Welfare*. MIT Press, 2003.
- [46] J. Nash. The bargaining problem. *Econometrica*, 18(2):155–162, 1950.
- [47] Y. Nesterov and V. Shikhman. Computation of Fisher–Gale equilibrium by auction. *Journal of the Operations Research Society of China*, 6(3):349–389, 2018.
- [48] N. Olver and L. A. Végh. A simpler and faster strongly polynomial algorithm for generalized flow maximization. *Journal of the ACM (JACM)*, 67(2):1–26, 2020.
- [49] K. Onaga. Dynamic programming of optimum flows in lossy communication nets. *IEEE Transactions on Circuit Theory*, 13:308–327, 1966.
- [50] A. Ruszczyński. *Nonlinear optimization*. Princeton university press, 2011.

- [51] H. Varian. Equity, envy and efficiency. *J. Econom. Theory*, 29(2):217–244, 1974.
- [52] V. Vazirani. The notion of a rational convex program, and an algorithm for the Arrow–Debreu Nash bargaining game. *Journal of the ACM (JACM)*, 59(2):1–36, 2012.
- [53] V. Vazirani and M. Yannakakis. Market equilibrium under separable, piecewise-linear, concave utilities. *J. ACM*, 58(3):10, 2011.
- [54] L. A. Végh. Concave generalized flows with applications to market equilibria. *Mathematics of Operations Research*, 39(2):573–596, 2014.